

**A WEB-BASED SERVICE MANAGEMENT SYSTEM WITH DATA
ANALYTICS USING MACHINE LEARNING ALGORITHMS FOR
SHOELOTSKEY VILLAMOR-PASAY**

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CHAPTER 1

INTRODUCTION

Small-scale shoe cleaning, restoration, and other service-oriented establishments play an important role in local economic activity, particularly in urban areas where they provide accessible products and services to the community. However, many of these businesses still depend on manual or semi-digital processes in managing their operations. The lack of integrated systems for handling transactions, inventory, and customer data often results in inefficiencies, inaccurate records, and limited capacity for business analysis. In an increasingly data-driven environment, the inability to utilize advanced technologies such as web-based platforms and data analytics puts small enterprises at a competitive disadvantage.

In the case of Ms. Aileen Rosales, owner of Shoelotskey, located in Villamor, Pasay City, operational challenges are evident in the management of services, sales transactions, and inventory. The current process relies heavily on manual recording, making it difficult to track customer purchases, monitor stock levels, and organize service-related data efficiently. This results in inconsistencies in records, delays in retrieving information, and challenges in identifying trends such as frequently purchased services or peak sales periods. Consequently, decision-making is often based on assumptions rather than data-driven insights, limiting the business's ability to optimize its operations and improve customer service.

Despite the growing availability of web-based management systems and machine learning technologies, there remains a significant research gap in developing integrated solutions specifically tailored for small-scale establishments like Shoelotskey. Existing systems are often designed for large enterprises, making them complex, costly, and unsuitable for businesses with limited resources

and technical expertise. Moreover, many studies focus either on basic automation or on advanced analytics independently, with minimal emphasis on combining service management systems and machine learning-based data analytics into a single, practical solution for small businesses.

To address these challenges, this study proposes the development of a web-based service management system integrated with data analytics using machine learning algorithms for Shoelotskey. The proposed system aims to automate core business processes such as service and sales management, inventory tracking, and report generation while incorporating machine learning techniques to analyze historical data and generate predictive insights. These insights may include sales trends and predicted service completion times. By implementing this system, the business is expected to enhance operational efficiency, improve accuracy, and support data-driven decision-making. Ultimately, this research seeks to provide an innovative yet practical solution that bridges the gap between advanced technology and the operational needs of small-scale establishments.

Problem Statement

Shoelotskey currently experiences specific operational challenges stemming from its reliance on manual processes. Based on preliminary observations and interviews with the business owner, the following problems have been identified:

Service records, customer information, and inventory data are recorded manually or maintained in separate files. This fragmentation makes data retrieval, updates, and monitoring time-consuming and error-prone.

The business lacks a centralized dashboard for monitoring daily operations. Business owner cannot easily access information about pending services, completed transactions, service demand trends, or sales performance.

Service completion date estimation depends entirely on staff judgment rather than historical data analysis. This results in inconsistent estimates and potential customer dissatisfaction.

Report preparation requires significant manual effort. Daily, weekly, and monthly, quarterly, and annually sales and performance summaries must be compiled manually, reducing administrative productivity.

According to preliminary data collected from Shoelotskey operations:

Average service processing delays due to manual status updates: 2-3 days

Current manual report preparation time: approximately 4-6 hours per month

Record retrieval time for customer history: 15-30 minutes per request

Monthly average transaction volume: 280-300 service orders

Without a structured system and data-driven approach, the business faces continued operational inefficiency and limited capacity for growth.

Research Questions

This study seeks to answer the following questions:

1. How can a web-based service management system be designed and implemented for Shoelotskey to centralize service transactions, customer records, and inventory monitoring?
2. How can machine learning algorithms be applied to predict service completion dates using historical service records and workload data?
3. How can data analytics features be integrated and evaluated to provide meaningful insights regarding sales trends, service demand, and business performance?
4. To what extent does the developed system meet ISO/IEC 25010 software quality standards in terms of:
 - 4.1 Functional suitability

- 4.2 Performance efficiency
- 4.3 Usability
- 4.4 Reliability
- 4.5 Security
- 4.6 Maintainability
- 4.7 Portability
- 5. What is the level of user acceptance of the developed system based on the Technology Acceptance Model (TAM) in terms of:
 - 5.1 Perceived usefulness
 - 5.2 Perceived ease of use
 - 5.3 Behavioral intention to use
 - 5.4 Actual system use
- 6. What improvements may be recommended for future enhancement of the system?

Project Context

Micro, small, and medium enterprises (MSMEs) play a significant role in economic development by generating employment opportunities, supporting local commerce, and providing accessible products and services to surrounding communities. In the Philippines, many small service-oriented businesses continue to rely on manual or semi-digital operational processes such as handwritten logbooks, spreadsheet records, and fragmented paper-based documentation. While these traditional practices may be manageable at a small scale, they often become inefficient as transaction volume increases. Common issues include delayed retrieval of records, inaccurate inventory counts, difficulty monitoring service progress, and limited access to timely business information.

The growing use of digital technologies has transformed how businesses manage operations, analyze performance, and improve customer service. Web-based information systems enable centralized data storage, real-time monitoring, automated reporting, and faster transaction processing. Likewise, business

analytics tools allow owners and managers to identify sales trends, customer demand patterns, operational bottlenecks, and performance opportunities. Businesses that fail to adopt these technologies may experience slower decision-making, higher administrative workload, and reduced competitiveness.

Shoelotskey, a shoe cleaning and restoration service business located in Villamor, Pasay City, owned by Ms. Aileen Rosales, currently experiences operational challenges associated with manual record management. Based on initial interviews and observation, service transactions, inventory usage, customer records, and service status updates are recorded manually or through isolated files. This setup makes it difficult to monitor pending and completed services, track frequently availed services, prepare sales summaries, and estimate completion schedules accurately. As a result, managerial decisions may rely more on assumptions than on organized operational data.

Another operational challenge involves predicting service completion dates. Customers commonly expect clear estimates regarding release schedules; however, manual estimation based only on staff experience may produce inconsistent results. Inaccurate estimates may lead to delays, customer dissatisfaction, and inefficient workload distribution. Predictive technologies such as machine learning can help address this issue by learning patterns from historical service data and generating more reliable completion estimates.

Although many studies discuss business automation, dashboards, and machine learning applications, limited research has focused on integrating these technologies into a single system specifically designed for micro service businesses such as shoe cleaning and restoration shops. Most available systems target large enterprises or broader retail environments, which may be too costly, complex, or unsuitable for small organizations with limited technical resources.

To address these concerns, this study proposes the development of a Web-Based Service Management System with Data Analytics Using Machine Learning

Algorithms for Shoelotskey Villamor-Pasay. The proposed system aims to centralize service transactions, inventory records, reporting processes, and performance monitoring while incorporating predictive analytics for service completion estimation. Through this system, Shoelotskey may improve operational efficiency, data accuracy, customer service responsiveness, and evidence-based decision-making.

Project Purpose and Description

This study aims to develop and evaluate a Web-Based Service Management System with Analytics and Machine Learning for Shoelotskey, a shoe cleaning and restoration service business located in Villamor, Pasay City. The study seeks to address operational issues such as manual record-keeping, fragmented files, delayed reporting, limited-service monitoring, and inconsistent estimation of service completion dates.

The proposed system is intended to centralize customer records, job orders, inventory data, service transactions, and reporting processes through a web-based platform accessible to authorized users. It includes modules for order entry, inventory monitoring, service status tracking, sales reporting, dashboard analytics, and machine learning-based completion date estimation.

The study will evaluate whether the developed system may contribute to measurable outcomes such as reduced processing time, faster report preparation, improved monitoring of service status, acceptable prediction accuracy, and higher user satisfaction.

System quality will be assessed using ISO/IEC 25010 standards, while user acceptance will be measured using the Technology Acceptance Model (TAM). The system is intended for internal use by Shoelotskey management and staff only.

This study may provide evidence on the practical application of web systems, analytics, and machine learning in small service-oriented businesses and

may serve as a reference for future researchers developing similar systems for MSMEs.

Conceptual Framework

The diagram illustrates an Input–Process–Output (IPO) model with a feedback loop designed for an efficient shoe service management system.

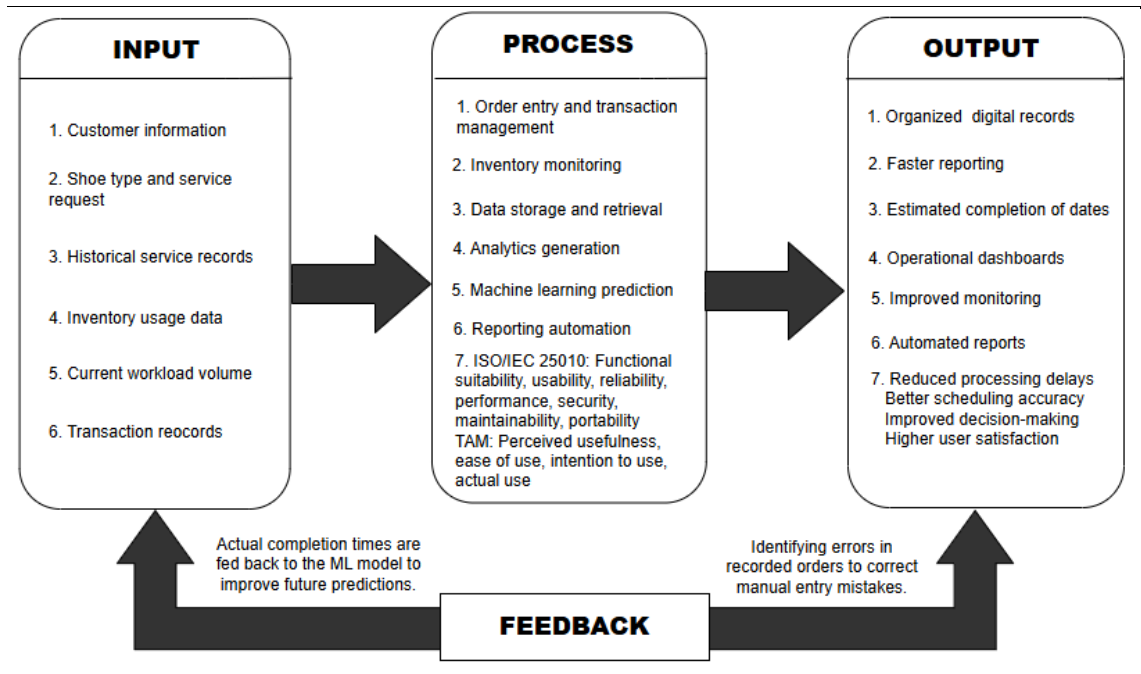


Figure 1: Conceptual Framework

The Input phase includes all the information the system needs to start working. This includes customer details, shoe types, and service requests. The system also takes in historical data, which are records of past services and current workloads. These inputs are necessary because they provide the raw data that the machine learning model needs to make its predictions.

The Process phase is where the actual work happens. The system manages orders, monitors inventory, and stores data securely. During this stage, the researchers apply the ISO/IEC 25010 standard to make sure the software is reliable, secure, and easy to maintain. This is also where the Random Forest algorithm analyzes the input data to predict completion dates. Furthermore, the

Technology Acceptance Model (TAM) is used here to ensure that the system is useful and easy to use for the staff, which encourages them to use the software daily.

The Output phase shows the final results of the system. These include organized digital records, automated sales reports, and accurate pick-up dates for customers. The main goal of the output is to reduce the delays found in the old manual system and to provide the owner with a clear dashboard to monitor the business.

The Feedback loop is a critical part of the framework. It allows the system to learn from its mistakes. For example, if a machine learning prediction was wrong, the actual completion time is fed back into the system to improve future accuracy. It also helps staff identify and correct manual entry errors. This continuous loop ensures that the Shoelotskey system becomes smarter and more accurate over time.

Objectives of the Study

This capstone project aims to design and develop the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms in order to improve operational efficiency, data accuracy, service monitoring, and managerial decision-making.

Specific Objectives

Specifically, the project aims to achieve the following:

1. Develop and implement a centralized digital job order entry and management system that captures customer information, service specifications, and transaction records, thereby reducing average service record retrieval time from 15-30 minutes to less than 5 minutes.
2. Implement and evaluate a machine learning model for predicting service completion dates with a target accuracy of 85% or higher compared to manual estimation methods.

3. Generate a real-time analytics dashboard that generates automated daily, weekly, monthly, quarterly, and annual reports, reducing manual report preparation time from 4-6 hours monthly to less than 1 hour.
4. Evaluate the effectiveness of the developed system based on the ISO/IEC 25010 software quality standards, specifically in terms of:
 - 4.1 Functional suitability
 - 4.2 Performance efficiency
 - 4.3 Usability
 - 4.4 Reliability
 - 4.5 Security
 - 4.6 Maintainability
 - 4.7 Portability

Additionally, the study will assess data accuracy and prediction reliability of the machine learning component.

5. Measure the level of user acceptance of the developed system among the staff and management of Shoelotskey using the Technology Acceptance Model (TAM), through a survey questionnaire measuring perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use.
6. Recommend possible system enhancements by identifying improvements and additional features that may be implemented by future researchers to further enhance system functionality, data analytics capabilities, and user experience.

Scope and Limitations

This study focuses on the design, development, and implementation of a Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey, a shoe cleaning and restoration service business located in Villamor, Pasay City. The system is intended to automate and improve the establishment's internal operations through a centralized and web-based platform accessible to authorized users such as the owner and staff.

The scope of the system includes user authentication and role-based access control, allowing secure login for administrators and staff with designated permissions. It includes a dashboard module that displays real-time operational summaries such as sales, job orders, completed services, pending transactions, and other business performance indicators through charts and reports.

The system also covers a job order summary management module that allows digital recording of customer information, shoe details, requested services, and transaction records. A service management module is included for maintaining service categories, pricing, and service descriptions offered by the business.

For inventory operations, the system includes an inventory management module that tracks materials used in services, updates stock quantities, stores inventory history, and provides low-stock notifications. A sales and reporting module is also included to generate daily, weekly, monthly, quarterly, and annual reports related to sales, expenses, and operational performance.

To improve scheduling and customer service accuracy, the system includes a machine learning prediction module that estimates service completion dates based on historical records and current workload conditions. A release calendar module is also integrated to monitor expected release dates, delayed services, and daily service schedules.

The system further includes search and record retrieval functions for customers, job orders, transactions, and service history, as well as responsive web access, allowing the system to be used on desktop computers, laptops, tablets, and mobile devices through a browser.

However, the system is limited to the internal operations of Shoelotskey only and is customized according to its current workflow and business requirements. It does not include a customer-facing online booking portal, mobile application, or self-service customer account access. External integrations such as online

payment gateways, SMS notifications, email alerts, accounting software, payroll systems, and third-party delivery services are not included in the present version.

The accuracy of the machine learning prediction module depends on the quality, consistency, and quantity of historical service data available. Prediction results may improve as more records are collected over time. The system also requires internet connection or local network access for full functionality, and offline mode is not supported in the current implementation.

Additionally, there are privacy and data handling limitations. The system requires careful management of customer personal data and maintains records of customer information and service history within its database. While the system is designed with secure access controls, data privacy measures are limited to user authentication and role-based access restrictions. The system does not include advanced encryption mechanisms for data at rest or comprehensive compliance with external data protection regulations. Therefore, data security relies primarily on institutional safeguards and proper user access management.

The effectiveness and reliability of study findings are also limited by the small sample size of respondents used during system testing and user acceptance testing phases. Due to the localized nature of the business and limited number of staff members and customers involved in the evaluation process, the generalizability of results may be constrained. Feedback and performance metrics obtained from this limited respondent pool may not fully represent scenarios in larger or more complex operational environments.

Multi-branch management, franchise operations, advanced enterprise analytics, and highly customized financial accounting features are beyond the scope of this study but may be considered for future enhancement.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents a thematic review of related literature that supports the development of a Web-Based Service Management System with Data Analytics using Machine Learning Algorithms. All studies cited in this review were published between 2022 and 2026 and were drawn from peer-reviewed journals, conference proceedings, and academic publications. The review of related literature is organized into five alphabetically arranged topics, each focusing on a key component of the study: Business Analytics and Sales Performance Dashboards, Centralized Digital Order Management and Data Accuracy, Machine Learning for Service Completion Time Prediction, Software Quality Evaluation Using ISO/IEC 25010 Standards, and the Technology Acceptance Model (TAM). Each section applies the Compare–Contrast–Connect approach by examining the similarities of previous studies, identifying their differences and limitations, and explaining how their findings contribute to and support the present research.

Business Analytics and Sales Performance Dashboards

Analytics dashboards have evolved from supplementary visualizations into essential decision-support tools embedded in web-based systems. Studies consistently show that well-designed dashboards improve operational visibility and speed up decision-making. Naprawski [1] demonstrated that web analytics enable managers to monitor service usage patterns and pinpoint improvement areas using real user data. Balolot et al. [2] similarly found that visual charts make performance metrics accessible to non-technical stakeholders, accelerating responses. These works converge on the value of intuitive presentation.

Yet important divergences emerge when moving beyond display to actionability. Dizon et al. [3] integrated analytics directly with scheduling

algorithms, creating proactive workload management rather than retrospective reporting. In contrast, Madlberger and Yesaulov [4] and Ramaswamy and Sankaran [5] offer broader industry reviews that highlight responsiveness gains but acknowledge implementation barriers in resource-constrained environments. Dela Vega et al. [6], the most contextually relevant, showed that even a small Philippine print shop could derive meaningful insights from basic regression-based analytics without heavy infrastructure. Canay and Kocabicak [7] focused on high-traffic enterprise frameworks, raising questions about scalability downward to micro-businesses.

A key limitation across several studies is the assumption of abundant data and technical support conditions rarely met by small trade shops like Shoelotskey. This study therefore adopts the pragmatic, low-overhead approach exemplified by Dela Vega et al. [6], incorporating simple daily/weekly/quarterly sales and service trend charts. By grounding the dashboard in the shop's own modest transaction volume, the system avoids the over-engineering critiqued in larger-scale frameworks while still delivering the responsiveness gains noted by Madlberger and Yesaulov [4] and Ramaswamy and Sankaran [5], directly supporting Research Question 3.

Centralized Digital Order Management and Data Accuracy

Manual logbooks remain common in small service businesses, yet they consistently produce errors, retrieval delays, and accountability gaps. The literature overwhelmingly endorses centralized digital systems as a remedy. Zeva [8] and Szedmak et al. [9] emphasize accessibility from any internet-connected device and measurable workflow improvements after replacing paper records. Coloma et al. [10], Almaceda Banawa and De Jesus [12], and Kristiani and Marcel [11] report concrete reductions in manual effort and error rates through structured data capture. Alduaij et al. [14] further link data quality to overall platform performance and user satisfaction.

Differences in context, however, reveal limitations. Most examples originate from schools, hospitals, government offices, or software firms—environments with higher transaction volumes and dedicated IT staff than a neighborhood shoe repair shop. Kristiani and Marcel [11] come closest by studying small firms, yet their focus remains on software development rather than craft-service operations. This contextual mismatch means general benefits do not automatically translate; small shops face unique data-scarcity and workflow constraints.

The Shoelotskey order-entry module directly applies the core principles of Zeva [8], Coloma et al. [10], and Almaceda Banawa and De Jesus [12] while adapting them to low-volume, manual-trade realities. By replacing the physical logbook with a single digital repository for customer, shoe, and service details, the system mitigates the very errors and delays highlighted in the literature. Alduaij et al. [14]’s emphasis on data accuracy also informs the ISO 25010 evaluation criteria used later, addressing Research Question 1.

Machine Learning for Service Completion Time Prediction

Accurate service completion estimates remain elusive when based on intuition alone. Machine learning offers a data-driven alternative, but applicability depends heavily on dataset size and model complexity. Caturkusuma et al. [15] achieved an R^2 of 0.98 with Random Forest on IT incident data, while Cribillero et al. [19] demonstrated >80% accuracy with Random Forest and SVM in small and medium Peruvian enterprises. Kurkute et al. [16] and Jevsejev and Bereisa [17] showed deep-learning (RNN/LSTM) and clustering models can reduce resolution times, and Ahmed et al. [18] reported up to 90% improvement in problem detection across reviewed studies. Gursoy and Kurt [20] envision ML becoming core to intelligent service systems.

Contradictions and limitations are evident. Deep-learning approaches [16] demand large time-series datasets that small shops cannot provide, whereas simpler tree-based models [15, 19] prove more feasible but may sacrifice some

predictive power. Ahmed et al. [18] explicitly note data-quality and organizational barriers that hinder adoption in practice. Most studies also draw from IT or manufacturing giants, not craft-service micro-businesses.

Shoelotskey's prediction model is deliberately trained on the shop's own historical records (shoe type, service requested, technician workload) using supervised methods akin to Caturkusuma et al. [15] and Cribillero et al. [19]. Evaluation employs the same MAE, RMSE, and R^2 metrics for transparency. This pragmatic, context-specific design overcomes the scalability critique leveled at enterprise-focused work and directly answers Research Question 2 while feeding both the analytics dashboard and ISO 25010 reliability assessment.

Software Quality Evaluation Using ISO/IEC 25010 Standards

ISO/IEC 25010 provides a standardized, multi-dimensional lens for software quality. Rojas et al. [13] confirmed its widespread adoption across domains through a bibliometric review of over 1,600 studies. Alduaij et al. [14] linked system, information, and service quality to user satisfaction via the DeLone and McLean model. Jacob [21], Rodriguez-Ortiz et al. [22], and Canay and Kocabicak [7] illustrate its application to ML-enhanced and analytics platforms, while Adewale et al. [23] and Deshpande [24] underscore the critical role of data preprocessing and usability—factors that technical metrics alone cannot capture. Kristiani and Marcel [11] reinforce that structured approaches improve reliability in small-firm settings.

While the standard is robust, prior applications often privilege large-scale or technical environments. Few studies combine ISO 25010 with user-centric evaluation in resource-limited Philippine MSMEs, and even fewer address the interplay of ML prediction accuracy and everyday usability. Deshpande [24] warns that technically sound systems can still fail if they ignore organizational fit.

The Shoelotskey system is therefore evaluated across functionality, data accuracy, prediction reliability, and user satisfaction using ISO/IEC 25010,

supplemented by staff feedback. This multi-angle approach, modeled on Rojas et al. [13], Alduaij et al. [14], and Jacob [21], ensures technical excellence does not come at the expense of practical adoption—directly addressing Research Question 4.

Technology Acceptance Model (TAM) for User Acceptance of Web-Based Service Systems

TAM remains the dominant framework for explaining technology adoption through Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). Recent Philippine and regional MSME studies [25–28] consistently show high PU and PEOU scores predict strong behavioral intention and actual usage, especially when shifting from manual processes. Usita et al. [25] and Estepa et al. [28] focused on food and catering micro-enterprises, while Reyes [26] and Adyaputra et al. [27] integrated TAM with ISO 25010, reporting dramatic reductions in processing time and errors alongside excellent acceptance scores.

Limitations persist: most evaluations target government, food, or inventory contexts rather than craft-service trades, and few examine systems that also embed ML prediction and analytics. Short post-implementation surveys may overlook sustained use in low-resource settings.

Shoelotskey therefore employs a dual ISO 25010 + TAM evaluation (following the mixed-methods instruments of Usita et al. [25], Reyes [26], and Adyaputra et al. [27]). This addresses both technical quality and genuine user perception among non-technical staff, fulfilling Research Question 4 and closing the user-centric gap evident in prior work.

Synthesis of the Reviewed Literature

Collectively, the five themes establish both the necessity and feasibility of an integrated web-based service management system for small Philippine service enterprises. Analytics dashboards [1–7], centralized digital records [8–12, 14], ML-

based prediction [15–20], ISO 25010 evaluation [7, 11, 13, 14, 21–24], and TAM [25–28] each supply proven components. Yet a critical reading reveals persistent gaps: most studies assume large datasets, dedicated IT support, or enterprise-scale operations. Small craft businesses—particularly Philippine shoe repair shops—remain underrepresented, and integrated solutions combining order management, ML prediction, analytics, and dual technical/user evaluation are virtually absent.

Prior work also surfaces contradictions worth noting. Enterprise frameworks [7, 16] deliver impressive performance but risk over-complexity for micro-businesses; simpler models succeed in small firms [6, 19] yet rarely scale to full-system integration. User-acceptance studies achieve high TAM scores but seldom test long-term adoption under real operational constraints. By designing a unified platform where clean order data directly trains a shop-specific ML model whose outputs feed intuitive dashboards—and rigorously evaluating the whole via ISO 25010 and TAM—this study not only builds on the literature but meaningfully extends it. It applies proven technologies to an understudied context, tests their synergistic effect in a low-volume, non-technical environment, and provides a replicable model for digital transformation in Philippine MSMEs. In doing so, it fills a clear void and offers practical, evidence-based guidance for similar small-service enterprises.

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CHAPTER 3

TECHNICAL BACKGROUND

Web-based information systems are widely used by service-oriented businesses to manage daily operations such as job orders, customer transactions, service records, scheduling, and report generation. Compared with manual logbooks or spreadsheet-based methods, web-based systems provide faster data access, centralized storage, real-time updates, and improved accuracy. In this study, a Web-Based Service Management System with Data Analytics using Machine Learning Algorithms was developed for Shoelotskey to improve service transaction management, monitor service progress, and generate operational reports through a centralized digital platform.

The proposed system follows a three-tier architecture, consisting of the presentation layer, application layer, and data layer. This architecture was selected because it separates the user interface, business logic, and database operations into organized components, making the system easier to maintain, troubleshoot, and expand. It also supports scalability, allowing additional modules or users to be accommodated in the future without major redesign.

The presentation layer is the front-end component accessed by users through standard web browsers. It was developed using React with TypeScript to ensure structured, reusable, and maintainable code. Vite was utilized as the build tool to provide fast development performance and optimized application loading. For the user interface, Tailwind CSS was implemented to create a clean, responsive, and user-friendly design across desktop and mobile devices. Lucide React was also integrated to provide consistent and professional icons throughout the system. Through this layer, users can encode transactions, update service status, search records, and generate reports efficiently.

The application layer contains the core system logic and server-side processing. It was developed using Python with the FastAPI framework, which was selected because of its high performance, lightweight structure, and ability to efficiently handle multiple user requests. FastAPI also supports RESTful API development, asynchronous processing, and seamless integration with analytical modules. This layer is responsible for user authentication, form validation, business rule processing, report generation, and communication between the front-end interface and the database.

The data layer uses PostgreSQL, a robust open-source relational database management system used for storing and managing operational data. PostgreSQL was selected because of its reliability, scalability, advanced querying features, and strong security controls. SQLAlchemy ORM was used to simplify database transactions and improve maintainability of database operations. The database includes structured tables such as users, customers, job orders, services, transactions, predicted schedules, and audit logs. Through normalization, indexing, and relational constraints, the system minimizes data redundancy while improving consistency and retrieval speed.

Different user roles were implemented to manage system access privileges. The owner or administrator has full access to dashboards, user management, reports, service monitoring, and analytics tools. Staff users are limited to operational functions such as recording customer transactions, updating job order status, and checking service schedules. This role-based access control enhances security and prevents unauthorized modifications.

The machine learning module is integrated into the system to predict estimated service completion timelines based on historical transaction data. The module was developed using the Scikit-Learn library in Python. Historical variables such as service type, workload volume, transaction date, and previous turnaround times were used as input features for prediction.

A Random Forest Regression model was selected because the required output is a continuous numerical value, specifically the predicted number of days or completion date. This algorithm was chosen because of its strong predictive accuracy, ability to handle nonlinear relationships, resistance to overfitting, and suitability for structured business datasets. Compared with simpler linear models, Random Forest can better capture patterns in service duration data.

Before model training, preprocessing techniques were applied, including removal of incomplete records, handling missing values, categorical encoding, and feature scaling when necessary. The dataset was divided into training and testing sets using an 80:20 ratio, where 80% of the records were used for training and 20% for model validation. Prediction performance was evaluated using Mean Absolute Error (MAE), which measures the average difference between predicted and actual completion dates. Lower MAE values indicate better predictive accuracy.

To ensure system security, user passwords are protected using the Bcrypt hashing algorithm. Additional safeguards include input validation, token or session-based authentication, controlled user permissions, and regular database backup procedures. These controls help protect confidential records and reduce risks related to unauthorized access, data loss, and misuse.

By integrating the React front-end, FastAPI backend, PostgreSQL database, and machine learning module, the proposed system supports both operational efficiency and predictive analytics. The selected technologies were chosen for affordability, maintainability, scalability, performance, and future expansion, making the solution practical for Shoelotskey's current operations while supporting long-term business growth.

Research and Development Methodology

This study adopted the Developmental Research Method combined with the Waterfall Software Development Life Cycle (SDLC) model. Developmental

research is appropriate because the study focuses on designing, developing, and evaluating a web-based service management system integrated with data analytics and machine learning algorithms.

The Waterfall model was selected to ensure a structured, systematic, and well-documented development process suitable for this capstone project. It follows a linear and sequential approach where each phase, Requirements Analysis, System Design, Implementation, Testing, Deployment, and Maintenance, is completed and formally documented before proceeding to the next, ensuring that requirements are clearly defined and approved early in the project.

Software Development Life Cycle Model

The development of the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms adopted the Waterfall Software Development Life Cycle (SDLC) model. This model was selected because it follows a systematic and phase-by-phase process that is appropriate for academic and capstone projects requiring clear documentation, defined outputs, and controlled implementation. Each development stage must be completed before proceeding to the next, allowing better requirement tracking, organized project management, and formal validation of deliverables (Sommerville, 2016; Pressman & Maxim, 2020).

The business operations of Shoelotskey, including job order processing, scheduling, service monitoring, and performance tracking, follow predictable and well-defined workflows. Since the system requirements were identified during the initial planning stage and were not expected to change significantly, the Waterfall model was considered more suitable than iterative approaches such as Agile, which are more effective for projects with rapidly changing requirements. For this study, a structured development process was more appropriate than frequent redesign cycles.

The model was also suitable because the system integrates data analytics and machine learning components that require organized preparation of datasets, database structures, and processing logic before implementation. Through the Waterfall approach, these technical foundations were properly designed during the earlier phases, reducing development errors and improving system consistency.

In terms of software quality assurance, the Waterfall model supports systematic testing and evaluation in every major phase of development. This strengthens compliance with ISO/IEC 25010 standards by enabling structured assessment of functionality, reliability, usability, maintainability, and performance efficiency.

Although the Waterfall model has limited flexibility when changes arise during later stages, this limitation was minimized through early requirement gathering, stakeholder interviews, and detailed planning conducted before development began. These measures helped ensure that the identified system requirements remained stable throughout the project lifecycle.

Figure 2 illustrates the Waterfall development workflow adopted in this study.

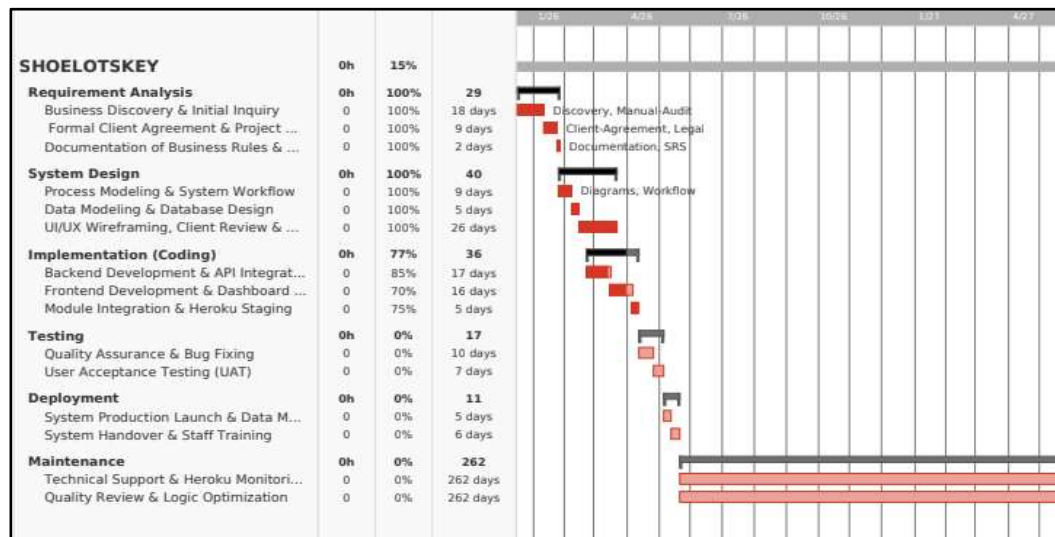


Figure 2: Gantt Chart

Showing the linear and sequential progression of system development phases, where each stage is completed and formally documented before moving

to the next, ensuring structured implementation, clear requirement definition, and controlled validation throughout the project lifecycle.

Waterfall Development Phases

The development of the Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms followed the Waterfall Software Development Life Cycle (SDLC) model. This model was adopted because it provides a structured and sequential approach in which each phase is completed before proceeding to the next. The methodology enabled the researchers to systematically gather requirements, design the system, develop the modules, test the functionalities, deploy the system, and perform post-deployment improvements for Shoelotskey.

Requirements Analysis

The requirements analysis phase focused on understanding the current operational procedures of Shoelotskey and identifying the problems encountered in managing services, inventory, sales, and records. The researchers conducted interviews, focus group discussions, and consultations with the business owner and staff to gather information regarding the existing manual processes, common difficulties, and desired system features.

Through this phase, the researchers identified the need for a centralized order management system, automated inventory monitoring, sales reporting, service tracking, and predictive service completion estimates. Functional and non-functional requirements were documented, including user roles, system security, usability, and reporting needs. The output of this phase was the System Requirements Specification (SRS), which served as the basis for the succeeding design phase.

System Design

During the system design phase, the researchers translated the identified requirements into a technical plan for system development. The overall architecture of the system was designed using a three-tier web-based structure consisting of the presentation layer, application layer, and database layer.

The researchers created design models such as the Context Diagram, Data Flow Diagrams (DFD), Entity Relationship Diagram (ERD), database schema, process flowcharts, and user interface prototypes. Navigation flow, dashboard layout, job order forms, inventory pages, and reporting interfaces were also designed. This phase ensured that the system structure, database relationships, and user interface were properly planned before coding began.

Implementation (Coding)

In the implementation phase, the researchers converted the approved system design into a working web-based application using appropriate web development technologies and database tools. The database was created and configured to store customer records, job orders, inventory data, service history, and transaction reports.

The researchers developed the major system modules, which included the Dashboard Module, Job Order Summary Management Module, Service Management Module, Release Calendar Module, Inventory Management Module, Sales Reporting Module, and User Management Module. The machine learning component was also integrated to predict estimated service completion dates based on historical records and workload data using the Random Forest Regressor model. This model was selected because of its ability to handle nonlinear relationships, improve prediction accuracy, and perform effectively using structured historical service data. All modules were connected to ensure proper data flow and system functionality.

Testing

The testing phase was conducted to verify whether the developed system met the specified requirements and performed correctly under actual use. The researchers carried out **functional testing** to ensure that each module operated as intended, such as saving records, generating reports, updating inventory, and displaying dashboards.

Integration testing was also performed to confirm that modules worked properly when connected to one another. In addition, **usability testing** was conducted with selected users to determine whether the interface was understandable and easy to use. Errors identified during testing were corrected, and system refinements were made to improve reliability, efficiency, and accuracy.

Deployment

After successful testing, the system was deployed for actual use in Shoelotskey. The researchers installed the system in the designated operational environment and configured user accounts for authorized personnel.

The owner and staff were oriented on how to use the system, including creating job orders, updating service status, checking reports, and monitoring inventory. During deployment, the researchers ensured that the system was accessible, properly configured, and ready for daily operations.

Maintenance

The maintenance phase focused on ensuring the continued performance and usability of the system after deployment. The researchers monitored the system for possible issues such as login errors, incorrect reports, or interface concerns encountered by users.

Minor revisions, bug fixes, and performance improvements were applied when necessary. User feedback was also gathered to identify future

enhancements such as additional reports, expanded analytics features, and further improvements to the prediction module. This phase helped maintain the long-term effectiveness and sustainability of the proposed system.

Machine Learning Based Service Completion Prediction

Machine learning was integrated into the proposed Web-Based Service Management System with Data Analytics for Shoelotskey to improve the accuracy of service completion time estimates and release dates. Although each shoe cleaning or repair service has a standard processing duration, actual completion time may vary depending on current shop workload, pending job orders, order priority (regular or rush), and service complexity. Through machine learning, the system provides more accurate and data-driven completion estimates.

The machine learning module was developed following the Waterfall Software Development Life Cycle (SDLC) to ensure systematic integration into the web-based platform. During the requirements analysis phase, approximately 500 historical completed job order records, service transactions, and turnaround time data were collected from Shoelotskey operations. Relevant variables included date received, date completed, service type, requested conditions, priority level, and workload volume.

For feature engineering, operational data were transformed into numerical variables such as the number of requested services, the number of reported shoe conditions, current workload volume, and priority level encoded using binary values, wherein rush orders were assigned a value of one while regular orders were assigned a value of zero. Additional adjustment factors were included to account for material type and condition severity. Date records were processed to compute actual turnaround time in days as the target variable.

During the design phase, the processed dataset was divided using an 80:20 train-test split, where 80% of the records were used for training and 20% for testing.

This method allowed the model to learn historical service patterns and evaluate predictive performance using unseen data. The trained model was then integrated into the application layer and connected to the job order form and service management page.

During the implementation phase, a regression-based approach was used because the objective of the study was to predict service completion time in days, which is a continuous numerical target variable. Among the regression algorithms considered, the Random Forest Regressor was selected over Linear Regression, Decision Tree, and XGBoost because it can effectively model nonlinear relationships among workload, priority level, and service complexity. It also reduces overfitting through ensemble learning, performs well on structured datasets from small and medium-sized businesses, and requires less parameter tuning than XGBoost. The predicted completion time was then used to calculate the estimated release date by adding the predicted completion days to the date received.

During the testing phase, model performance was evaluated using the Mean Absolute Error (MAE) metric, which measures the average difference between predicted and actual completion time. Lower MAE values indicate better prediction accuracy and more reliable release date estimates. Through this integration, the proposed system supports data-driven monitoring of service completion, improves scheduling efficiency, enhances customer satisfaction through accurate release dates, and strengthens operational decision-making for Shoelotskey.

Methodology Justification using Waterfall Model

The Waterfall development model was selected as the methodology for this study because the system requirements were clearly defined and stable at the start of the project. The proposed Web-Based Service Management System for Shoelotskey was developed based on identified business processes, documented

service records, and user requirements gathered through interviews and focus group discussions. Since the scope and functionality of the system were well understood early in the study, the sequential nature of the Waterfall model was appropriate.

The Waterfall model follows a structured and systematic approach that includes requirements analysis, system design, development, testing, implementation, and maintenance. This step-by-step process allowed the researchers to focus on completing each phase thoroughly before moving to the next, reducing errors and ensuring that the system design aligned with the actual operational needs of the business. It also made documentation easier to manage, which is essential for academic research and capstone projects.

Furthermore, the Waterfall model was suitable for integrating the machine learning component of the system. The availability of historical service data made it possible to define the machine learning requirements during the early stages of development. This allowed the researchers to design and implement the service completion date prediction model during the development phase and evaluate it during testing. Overall, the Waterfall methodology provided a clear development structure, supported proper documentation, and ensured that the proposed system met the objectives of the study within the given timeframe.

System Development Framework

The system development framework follows a Three-Tier Architecture, integrating the presentation layer, application layer, and data management layer to support efficient system operation and organized data processing. The development workflow includes requirements gathering, system design, module development, integration testing, and system validation to ensure that the proposed platform meets the operational needs of Shoelotskey.

The system was developed using modern web technologies, including server-side scripting, relational database management systems, and client-side

interface frameworks to support web-based access and centralized service management. The layered architecture allows the system to separate user interface functions, system processing logic, and database management, improving maintainability, scalability, and system performance.

Machine learning components are integrated within the application layer to support predictive analytics and service completion estimation. These components analyze historical service records and job order data to generate predicted completion times and identify service trends, assisting administrators in improving operational planning and decision-making.

Analysis of the Existing System

The analysis of the existing system examines Shoelotskey's current operational procedures for managing service transactions, job orders, and business records. Understanding the current workflow is essential in identifying limitations that affect service monitoring, data management, and overall business efficiency. This analysis focused on how service requests are recorded, how service completion is monitored, and how transactions are documented within the organization.

Currently, Shoelotskey relies on manual recording procedures and basic spreadsheet tools to manage daily operations. Service transactions and job orders are encoded manually, while service progress updates are monitored through handwritten notes or verbal coordination among staff. Although these methods allow basic record keeping, they do not provide an integrated platform for storing, updating, and retrieving operational data efficiently.

Based on observations and interviews, several operational issues were identified. Retrieving previous customer or service records often requires searching multiple files or spreadsheets, resulting in an estimated delay of 5 to 10 minutes per transaction inquiry. Manual encoding also increases the possibility of data entry errors, with approximately 2 to 3 record corrections per day caused by

duplicate entries, incomplete details, or inaccurate information. In addition, monitoring service completion status becomes difficult during busy periods, causing delayed updates and occasional miscommunication between staff and customers.

The current system also has limited capability for generating reports and analyzing business performance. Daily transactions, service volume, turnaround time, and customer trends must be summarized manually, consuming additional staff time and reducing management efficiency. Since historical records are not centralized, management has difficulty identifying service demand patterns and making timely operational decisions.

These findings indicate the need for a more efficient and automated system that can centralize records, improve service monitoring, reduce manual errors, and support data-driven decision-making for Shoelotskey.

Current Service Management Procedures

The current service management process at Shoelotskey primarily relies on manual recording of service transactions and job orders. When customers request services, staff members manually record service details using paper-based forms or spreadsheet files. Staff members manually enter information such as customer name, service type, service cost, and service date manually and store it in separate records.

After a service request is recorded, the job order is processed and performed in accordance with the requested service. Once the service is completed, staff members manually update the service status and record the transaction details in the existing record-keeping system. However, this process often requires staff members to review multiple documents or files to verify whether a service request has already been completed or is still pending.

Monitoring service schedules and completed services is also performed manually. Staff members must manually track which services have been released

or completed, which can delay record updates. This manual monitoring process becomes more challenging as the volume of service transactions increases.

Sales monitoring and reporting are conducted by manually reviewing recorded transactions. Staff members must compile service records and calculate totals manually to generate reports for business monitoring. This process requires considerable time and effort, particularly when reviewing service performance or preparing transaction summaries. As a result, the existing system limits Shoelotskey's ability to quickly access operational information and evaluate business performance.

Limitations and Problem Analysis

Analysis of current service management procedures revealed several operational and administrative limitations affecting the efficiency and reliability of business operations. One of the primary issues is the heavy reliance on traditional record-keeping, which increases staff workload and the risk of data entry errors. This approach also reduces efficiency in tracking service transactions and updating service records.

Another significant limitation is the difficulty in monitoring job orders and service completion. Since service status updates are recorded manually, staff members must check multiple records to determine whether services are completed, pending, or scheduled for release. This process makes it difficult to track service progress in real time and may delay service record updates.

The absence of a centralized database also presents a major challenge. Service records, job orders, and sales information are stored in separate documents or spreadsheet files, making information retrieval inefficient. When the business owner needs to review service history or analyze operational performance, staff members must search through multiple records to gather the required data.

In addition, the current system lacks analytical capabilities to identify trends in service demand, customer preferences, and service completion performance. Without automated data analysis tools, the organization cannot effectively use operational data for informed decision-making. These limitations highlight the need for an integrated and automated system that improves data management, monitoring, and overall operational efficiency.

Requirements Gap Analysis

A requirements gap analysis was conducted to determine the differences between Shoelotskey's existing manual service management procedures and the functional capabilities required for an efficient automated system. The analysis identifies missing features, process limitations, and operational needs necessary to improve the current workflow.

The findings show that the current system lacks a centralized platform for recording and managing service transactions. Service records are stored in separate logbooks and spreadsheet files, making it difficult for staff to access updated information and monitor job orders in real time. This results in fragmented records, slower transaction processing, and limited data accuracy.

Another major gap is the absence of automated tools for service monitoring and scheduling. Staff members manually check service progress and update completion status, which increases the possibility of delayed updates, missed schedules, and inefficient workflow coordination.

The existing process also lacks automated report generation and data analytics capabilities. Reports on daily transactions, service volume, and turnaround time must be prepared manually, consuming time and limiting management's ability to evaluate performance trends and make timely decisions.

These identified gaps indicate the need for a web-based service management system that can centralize records, automate service monitoring,

generate real-time reports, and provide analytical insights for improved decision-making.

Current State	Required State
Service records stored in separate logbooks and spreadsheets	Centralized digital database for all service records
Manual monitoring of job orders and completion status	Real-time tracking of job orders and service progress
Manual scheduling and status updates	Automated scheduling and notification system
Reports prepared manually	Automated report generation with dashboards
Limited historical data analysis	Data analytics for trends, demand forecasting, and performance evaluation
High risk of duplicate or incomplete entries	Accurate, validated, and consistent data management

Alternative Solutions and Feasibility Analysis

To address the identified limitations of the current system, several alternative solutions were evaluated to improve Shoelotskey's service management operations. Each option was assessed using key criteria such as functionality, scalability, cost efficiency, automation capability, reporting features, and compatibility with business operations. The purpose of this analysis was to determine the most suitable and sustainable solution for the organization.

The first alternative was to continue using the existing manual record-keeping process with minor improvements in documentation practices. This option requires minimal cost and no major system changes; however, it offers limited improvement in operational efficiency, data accessibility, and service monitoring. Manual processes also remain prone to delays, missing records, and encoding errors.

The second alternative was the use of spreadsheet-based management tools. Spreadsheet software can improve data organization, record storage, and basic reporting compared to purely manual methods. However, it still lacks real-time automation, centralized multi-user access, advanced analytics, and efficient

transaction monitoring. As service transactions increase, spreadsheet files may become difficult to maintain and more susceptible to version control issues.

The third alternative was the development of a custom web-based service management system specifically designed for Shoelotskey. This solution offers the highest level of flexibility because it can be tailored to the organization's workflow and operational requirements. It also supports centralized database management, real-time service monitoring, automated reporting, user access control, and future integration of data analytics and machine learning features.

Comparative Evaluation of Alternatives

Criteria	Weight	Manual Process	Spreadsheet System	Custom Web-Based System
Functionality	30%	2	3	5
Scalability	20%	1	2	5
Cost Efficiency	15%	5	4	3
Automation Capability	20%	1	2	5
Reporting & Analytics	10%	1	3	5
Compatibility with Operations	5%	3	4	5

Rating Scale: 1 = Poor, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent

Weighted Decision Matrix

Alternative	Weighted Score
Manual Process	2.00
Spreadsheet System	2.75
Custom Web-Based System	4.70

Based on the comparative evaluation, the **custom web-based service management system** achieved the highest weighted score. Although it requires higher initial development cost, it provides the best long-term value through improved efficiency, centralized operations, automation, scalability, and enhanced decision-making capabilities. Therefore, it was determined to be the most appropriate solution for Shoelotskey.

Proposed Technology Alternatives

Several technology solutions were evaluated to address the operational limitations identified in the existing service management procedures. Each technology alternative was examined based on its ability to support automation, centralized data management, scalability, and operational efficiency.

One alternative considered was enhancing the existing spreadsheet-based record management system. While spreadsheets can improve data organization, they still rely heavily on manual data entry and lack automated monitoring capabilities. Additionally, spreadsheet systems do not provide integrated dashboards, centralized database management, or advanced analytics features.

Another alternative was to adopt commercial service management platforms available on the market. These platforms offer automated features, including service tracking, reporting tools, and data storage. However, many commercial solutions require subscription fees and provide limited customization options. These limitations may prevent the system from fully addressing Shoelotskey's unique operational requirements.

The most suitable solution identified was the development of a custom web-based service management system. This approach enables the system to be specifically designed to meet the organization's operational needs. The proposed system includes modules for job order summary management, service completion tracking, sales reporting, dashboard monitoring, and user management. Additionally, the system integrates data analytics and machine learning capabilities to analyze service completion data and identify operational trends.

By implementing a custom web-based solution, Shoelotskey can improve service monitoring, centralize business records, and enhance decision-making through data-driven insights.

Feasibility Study

The feasibility study evaluates the practicality and sustainability of developing and implementing the proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithms for Shoelotskey. The objective of this assessment is to determine whether the proposed system can be successfully developed using available resources while meeting the organization's operational requirements. The feasibility study covers four major dimensions: technical, operational, economic, and schedule feasibility.

Technical Feasibility

Technical feasibility examines whether the required hardware, software, and technical resources are available to support system development and deployment. Shoelotskey currently has access to one computer unit, standard printer, and stable internet connectivity with an estimated average speed of 50 Mbps, which is sufficient for operating a web-based system.

The proposed system will be developed using widely available technologies such as Python, PostgreSQL, Tailwind CSS, JavaScript, and open-source development frameworks. These technologies are commonly used for small- to medium-scale business systems because they are cost-effective, reliable, and easy to maintain.

The system will use a three-tier architecture consisting of the presentation layer, application layer, and database layer. This structure improves maintainability, security, and system performance. The centralized database will store service records, job orders, transactions, and user accounts in a single platform.

Potential risks include increased data volume, server downtime, and database management complexity. However, these risks can be mitigated through regular backups, optimized database design, and preventive maintenance. Based

on available infrastructure and accessible technologies, the proposed system is considered technically feasible.

Operational Feasibility

Operational feasibility assesses whether the proposed system fits Shoelotskey's daily workflow and whether users can effectively adopt the system. At present, service transactions, job orders, and reports are managed manually using notebooks and spreadsheets, resulting in delays, duplicate entries, and difficulty tracking service status.

Based on interviews and observations, 5 staff members expressed willingness to use a digital system if basic training is provided. Staff members identified faster record retrieval, easier monitoring, and automated reports as major expected benefits.

The proposed system will allow users to record transactions, update job order status, track service completion, and generate reports through a user-friendly web interface accessible via standard browsers. Since the system improves existing procedures rather than replacing them entirely, resistance to adoption is expected to be minimal. Therefore, the proposed system is considered operationally feasible.

Economic Feasibility

Economic feasibility evaluates whether the expected benefits justify the cost of development and implementation. Since the system will use existing computer units and open-source software, implementation cost is limited mainly to development, internet usage, and maintenance expenses.

The current manual process requires significant administrative time for recording transactions and preparing reports. Based on observation, staff spend approximately 2 hours daily on manual encoding, searching records, and preparing summaries. With automation, these tasks are expected to be reduced to 30 minutes daily, saving around 30 to 35 staff hours per month.

Tangible benefits include reduced paper usage, faster report generation, improved record accuracy, and lower administrative workload. Intangible benefits include better customer service, improved monitoring of operations, and enhanced decision-making through data analytics. Given the manageable cost and long-term efficiency gains, the proposed system is considered economically feasible.

Schedule Feasibility

Schedule feasibility evaluates whether the proposed system can be developed and implemented within the planned project timeline. The development of the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms will follow the Waterfall Software Development Life Cycle (SDLC), which provides a structured and sequential approach to project execution. This methodology ensures that each phase is completed before proceeding to the next stage, allowing better control of project activities, deliverables, and deadlines.

The estimated duration for completing the project is approximately five months. The first phase, requirements analysis, is expected to take three weeks and will focus on gathering system requirements, identifying user needs, and analyzing existing business processes. This will be followed by the system design phase, which is estimated to take four weeks and will cover database design, interface layouts, and overall system architecture.

The system development phase is projected to require eight weeks, during which the core modules, database functions, and user interfaces will be coded and integrated. After development, testing and debugging activities are expected to take three weeks to verify system functionality, identify errors, and ensure that all requirements are met. The final phase, deployment and user training, is estimated to take two weeks and will involve installing the system, orienting users, and preparing the organization for operational use.

Possible scheduling risks include delays in coding, technical issues during testing, and revisions in user requirements. However, these risks can be minimized through effective planning, clear task allocation among team members, regular progress monitoring, and adherence to the proposed schedule. Based on the defined scope, available resources, and structured development methodology, the proposed system is considered schedule feasible and can be completed within the planned timeframe.

Cost–Benefit Analysis

Overview

The Cost–Benefit Analysis (CBA) evaluates the financial viability of implementing the proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithms for Shoelotskey Villamor-Pasay. The purpose of this analysis is to determine whether the expected operational and financial benefits of the system justify the total cost of development, deployment, and maintenance. By automating manual processes, centralizing records, and improving reporting efficiency, the proposed system is expected to provide measurable long-term value to the organization.

A. Cost Analysis

The cost analysis includes the estimated expenses for documentation materials, system development, operational needs, implementation, and web hosting.

1. Documentation and Office Supply Costs

Item	Quantity/Assumption	Unit Cost (₱)	Estimated Cost (₱)
Bond Paper A4	2 reams	350	700
Long Folder	10 pieces	10	100
Ballpen (Tri Quick Dry Gel 0.5)	5 pieces	20	100

Pentel Pen (Mark-4-all Permanent)	2 pieces	50	100
Logbook	2 pieces	150	300
Correction Tape (Getwin 6m)	2 pieces	27	54
Clip	10 pieces	8	80

Subtotal – Documentation and Office Supplies: ₱1,434

2. System Development Costs

Category	Description	Estimated Cost (₱)
Software and Development Tools	Use of development and testing tools	5,000
Programmer/Developer Effort	System design, coding, integration, and testing	25,000
Database Design and Setup	Database structure and configuration	8,000
Machine Learning Integration	Analytics dashboard and ML model integration	10,000

Subtotal – Development Costs: ₱48,000

3. Operational and Implementation Costs

Category	Description	Estimated Cost (₱)
Internet and Utilities	Internet and electricity during development/testing	3,000
Maintenance and Minor Revisions	Initial maintenance and adjustments	5,000
Training and Orientation	Basic user orientation for owner and staff	3,000

Subtotal – Operational and Implementation Costs: ₱11,000

4. Hosting and Deployment Costs

Category	Description	Estimated Cost (₱)
Web Hosting	Annual hosting subscription for live deployment	3,500
Domain Name	Annual domain registration	1,000
Deployment/Server Configuration	Initial setup and configuration of the hosting environment	2,000

Subtotal – Hosting and Deployment Costs: ₱6,500

The estimated implementation cost includes documentation materials, system development, operational expenses, maintenance, and web hosting requirements. Existing computer units will be utilized, reducing additional hardware expenses.

Cost Category	Estimated Cost (₱)
Documentation and Office Supplies	1,434
System Development	48,000
Operational and Implementation Costs	11,000
Hosting and Deployment Costs	6,500
Total Initial Project Cost	66,934

To evaluate long-term sustainability, annual operational and maintenance expenses were projected at **₱11,000 per year**.

Total Cost Over Three Years

$$\text{Total Cost} = \text{Initial Cost} + (\text{Annual Maintenance} \times 3)$$

$$= 66,934 + (11,000 \times 3)$$

$$= 66,934 + 33,000$$

$$= \text{P}99,934$$

Benefit Analysis

The system is expected to generate measurable annual benefits through improved efficiency, reduced errors, and better reporting processes.

Tangible Benefit	Estimated Annual Value per Year (P)
Reduced Manual Recording Errors	12,000
Faster Transaction Processing	15,000
Improved Report Generation	10,000
Better Service Monitoring	8,000
Data-Driven Decision Support	10,000
Total Annual Tangible Benefits	55,000

Intangible Benefit	Description	Estimated Value (P)
Improved organization of business records	Centralizes records for easier storage, tracking, and retrieval	Qualitative
Better monitoring of daily operations	Helps track services, transactions, and work progress more efficiently	Qualitative

Faster access to reports and service information	Allows quick viewing of reports and service details when needed	Qualitative
Stronger decision-making through analytics outputs	Provides useful insights that support better business decisions	Qualitative
Improved customer satisfaction through timely service handling	Enables faster and more accurate service processing for customers	Qualitative

Total Benefits Over Three Years

Total Benefits = 55,000 × 3 = ₱165,000

In addition to measurable benefits, the system also provides intangible advantages such as improved customer satisfaction, better record organization, faster access to information, and enhanced management decision-making.

Financial Evaluation

Net Benefit

Net Benefit = Total Benefits – Total Costs

= 165,000 – 99,934

Net Benefit = ₱65,066

Return on Investment (ROI)

ROI = (Net Benefit / Total Cost) × 100

= (65,066 / 99,934) × 100

ROI = 65.11%

Benefit–Cost Ratio (BCR)

$$BCR = (Total\ Benefits / Total\ Costs)$$

$$= (165,000 / 99,934)$$

$$BCR = 1.65 : 1$$

This means that for every **₱1.00** invested in the project, Shoelotskey is expected to gain approximately **₱1.65** in return over a three-year period.

Conclusion

The Cost–Benefit Analysis indicates that the proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithms is a financially feasible investment for Shoelotskey. With a total three-year cost of ₱99,934 and projected benefits of ₱165,000, the system yields a net benefit of ₱65,066, an ROI of 65.11%, and a Benefit–Cost Ratio of 1.65:1. These results demonstrate that the project can provide sustainable economic value through improved operational efficiency, reduced administrative workload, and better decision-making capabilities.

Development Cost (Waterfall Development Method)

The proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithms will be developed using the Waterfall Software Development Life Cycle (SDLC) model. This approach divides the project into sequential phases: Requirements Analysis, System Design, System Development, Testing and Deployment, System Evaluation, and Maintenance Support. Each phase must be completed before proceeding to the next stage, allowing better control of project scope, cost, and deliverables.

To estimate development cost realistically, updated labor rates with conservative assumptions were applied. These rates are based on entry-level

project compensation suitable for student-led or small-scale business system development.

Labor Cost Assumptions

Role	Main Responsibility	Hourly Rate (P)
Project Manager	Planning, supervision, coordination, approval	250
System Analyst	Requirements analysis, documentation, evaluation	220
Developer / Programmer	Coding, database setup, integration, maintenance	200
QA Tester	Testing, validation, user acceptance support	180

Cost Formula Used

Cost = No. of Persons × Hours Worked × Hourly Rate

Phase 1 – Requirements Analysis

This phase includes gathering requirements, stakeholder consultation, workflow analysis, and documentation.

Activity	Role	Persons	Hours	Rate (P)	Computation	Cost (P)
Project Planning Meeting	Project Manager	1	6	250	$1 \times 6 \times 250$	1,500
Business Process Analysis	System Analyst	1	10	220	$1 \times 10 \times 220$	2,200
Technical Consultation	Developer	1	5	200	$1 \times 5 \times 200$	1,000
Initial Validation	QA Tester	1	4	180	$1 \times 4 \times 180$	720

Total Phase 1 Cost

$1,500 + 2,200 + 1,000 + 720 = \text{P}5,420$

Phase 2 – System Design

This phase includes database design, interface planning, architecture design, and module structure.

Activity	Role	Persons	Hours	Rate (P)	Computation	Cost (P)
System Architecture Planning	Project Manager	1	8	250	$1 \times 8 \times 250$	2,000
Database Design	System Analyst	1	10	220	$1 \times 10 \times 220$	2,200
UI/UX Design Review	Developer	1	8	200	$1 \times 8 \times 200$	1,600
Design Validation	QA Tester	1	4	180	$1 \times 4 \times 180$	720

Total Phase 2 Cost

$2,000 + 2,200 + 1,600 + 720 = \text{P}6,520$

Phase 3 – System Development

This phase includes coding, front-end and back-end integration, database implementation, and machine learning module setup.

Activity	Role	Persons	Hours	Rate (P)	Computation	Cost (P)
Development Supervision	Project Manager	1	10	250	$1 \times 10 \times 250$	2,500
Database & Logic Review	System Analyst	1	8	220	$1 \times 8 \times 220$	1,760
Front-End Development	Developer	1	35	200	$1 \times 35 \times 200$	7,000
Back-End Development	Developer	1	40	200	$1 \times 40 \times 200$	8,000
ML Module Integration	Developer	1	15	200	$1 \times 15 \times 200$	3,000
Code Inspection	QA Tester	1	10	180	$1 \times 10 \times 180$	1,800

Total Phase 3 Cost

$2,500 + 1,760 + 7,000 + 8,000 + 3,000 + 1,800 = \text{P}24,060$

Phase 4 – Testing and Deployment

This phase includes system testing, bug fixing, deployment, and user training.

Activity	Role	Persons	Hours	Rate (P)	Computation	Cost (P)
Deployment Oversight	Project Manager	1	6	250	$1 \times 6 \times 250$	1,500
Functional Review	System Analyst	1	8	220	$1 \times 8 \times 220$	1,760
Bug Fixing	Developer	1	15	200	$1 \times 15 \times 200$	3,000
Full Testing & UAT	QA Tester	1	18	180	$1 \times 18 \times 180$	3,240

Total Phase 4 Cost

$$1,500 + 1,760 + 3,000 + 3,240 = \text{P}9,500$$

Phase 5 – System Evaluation

This phase includes post-deployment review, feedback gathering, and system performance assessment.

Activity	Role	Persons	Hours	Rate (P)	Computation	Cost (P)
Project Review	Project Manager	1	5	250	$1 \times 5 \times 250$	1,250
Performance Evaluation	System Analyst	1	8	220	$1 \times 8 \times 220$	1,760
Minor Adjustments	Developer	1	10	200	$1 \times 10 \times 200$	2,000
Feedback Testing	QA Tester	1	6	180	$1 \times 6 \times 180$	1,080

Total Phase 5 Cost

$$1,250 + 1,760 + 2,000 + 1,080 = \text{P}6,090$$

Phase 6 – Maintenance and Support

This phase includes corrective maintenance, updates, and monitoring for the first three months.

Activity	Role	Persons	Hours	Rate (P)	Computation	Cost (P)
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Maintenance Supervision	Project Manager	1	4	250	$1 \times 4 \times 250$	1,000
System Monitoring	System Analyst	1	6	220	$1 \times 6 \times 220$	1,320
Bug Fixes / Updates	Developer	1	15	200	$1 \times 15 \times 200$	3,000
Regression Testing	QA Tester	1	8	180	$1 \times 8 \times 180$	1,440

Total Phase 6 Cost

$1,000 + 1,320 + 3,000 + 1,440 = \text{P}6,760$

Overall Waterfall Development Cost Summary

Phase	Total Cost (P)
Phase 1 – Requirements Analysis	5,420
Phase 2 – System Design	6,520
Phase 3 – System Development	24,060
Phase 4 – Testing and Deployment	9,500
Phase 5 – System Evaluation	6,090
Phase 6 – Maintenance and Support	6,760
Total Development Cost	P58,350

Conclusion

The estimated **P58,350 total development cost** reflects a realistic and conservative budget for a small-scale academic or startup business system using the Waterfall SDLC model. The largest portion of the budget is allocated to the **System Development Phase**, as coding, database integration, and machine learning implementation require the most time and technical effort. This analysis shows that the proposed system can be developed at a manageable cost while maintaining quality, functionality, and long-term usability.

System Requirements Specification

The System Requirements Specification details the operational capabilities and technical integrations of the proposed Web-Based Service Management System. Developed specifically for Shoelotskey in Barangay 183 Villamor Pasay, the system integrates modern data analytics and machine learning algorithms to

transition the business from a manual ledger process into a fully digitized platform. The primary objective is to automate the recording of customer orders, calculate exact financial reports, and formulate accurate predictions regarding service completion timelines. By utilizing the Python Scikit-Learn library, the system actively analyzes historical processing times to provide the business owner with reliable service completion forecasts. The requirements specification serves as a guide for system development, ensuring that the implemented system meets Shoelotskey's operational needs while maintaining system reliability, security, and efficiency.

Stakeholder Identification and User Roles

The proposed system involves two primary stakeholders who interact with it according to their operational responsibilities. Identifying user roles ensures that system functionalities align with the business workflow while maintaining proper access control and system security.

The Owner serves as the primary administrator of the system and is responsible for monitoring overall service operations and business performance. The owner has full access to the system features, including viewing operational dashboards, generating sales reports, managing service information, analyzing service performance trends and managing the users. Through the system dashboard, the owner can review service completion statistics, transaction summaries, and analytics generated from historical service records.

The Staff users are responsible for managing the business's daily service operations. Staff members use the system to record customer service requests, create job orders, update service status, and monitor service completion. Staff users ensure that service transactions are accurately recorded in the system and that job order information is properly maintained.

The implementation of role-based access control ensures that system functions are accessible only to authorized users according to their responsibilities, thereby improving system security and operational efficiency.

Functional Requirements

The functional requirements describe the core system operations that support Shoelotskey's service management activities. These requirements define how users interact with the system and how the system processes service-related data.

The system shall provide secure user authentication, allowing authorized users to access the system using login credentials. This feature ensures that only the owner and staff members can access system functionalities based on their assigned roles.

The system shall provide a job order summary management function that allows staff members to record service requests, store customer information, and maintain job order details. Staff users shall be able to update service status and record service completion once the requested service has been performed.

The system shall include a service management module that allows the owner to manage available services and update service pricing information. This functionality ensures that service details are maintained accurately within the system database.

The system shall provide a dashboard module that displays summarized operational information, including the number of completed services, pending job orders, and sales performance indicators. This dashboard enables the owner to monitor business activities in real time.

The system shall include a released calendar module that allows users to track scheduled service releases and monitor service completion timelines.

The system shall generate sales reports that summarize service transactions and revenue performance. These reports allow the owner to evaluate business performance and identify trends in service demand.

The system shall integrate machine learning-based data analytics to analyze historical service records and identify patterns related to service completion, transaction frequency, and operational performance. The generated insights support better monitoring and decision-making for the business.

Non-Functional Requirements

The non-functional requirements define the quality attributes necessary to ensure that the proposed system operates efficiently, securely, and reliably. These requirements are based on recognized software quality characteristics such as security, usability, reliability, performance efficiency, and maintainability.

The system shall ensure security by implementing authentication mechanisms and role-based access control to prevent unauthorized access to system data. User login credentials will be required before accessing system features.

The system shall maintain usability by providing a user-friendly interface that allows both the owner and staff members to easily navigate system modules and perform operational tasks with minimal training.

The system shall ensure reliability by maintaining stable system performance during service transactions and database operations. Data backup mechanisms will be implemented to prevent data loss.

The system shall support performance efficiency by enabling fast retrieval of service records, efficient data processing, and quick generation of operational reports and dashboards.

The system shall also support maintainability by implementing a modular system architecture and a structured database design that allows developers to update, modify, and maintain the system efficiently.

Proposed System Architecture and Design

The proposed system architecture defines the overall structure of the Web-Based Service Management System for Shoelotskey. The architecture is designed to support centralized service data management, secure system access, and efficient processing of service transactions.

The system adopts a Three-Tier Architecture, which separates system components into three layers: the presentation layer, application layer, and data management layer. This layered structure improves system scalability, maintainability, and performance by separating user interface functions, application processing logic, and database operations.

Through this architecture, users interact with the system through web-based interfaces, while system processing and data management operations are handled within the application and database layers.

System Architecture Overview

The system architecture provides a structured framework that supports efficient system operations and centralized data processing. The presentation layer is the user interface through which system users interact with the platform via web browsers and developed using React and Vite. This layer provides access to modules including the dashboard, job order management, service management, sales reporting, the released calendar, and user management.

The application layer serves as the system's core processing component. It is built using Python with the FastAPI framework. It processes user requests, manages system logic, handles service completion updates, and performs machine learning analysis for data analytics. This layer ensures that operational

processes are executed efficiently before data is stored or retrieved from the database.

The data management layer manages the centralized database that stores service records, job orders, transaction data, and user account information. It uses a cloud-based PostgreSQL database to securely store all operational and transactional data of the system. This layer ensures secure storage, efficient retrieval, and consistent management of operational data required for reporting and analytics.

System Components and Modules

The proposed Web-Based Service Management System with Data Analytics for Shoelotskey comprises several functional modules designed to support the business's daily service management operations. Each module performs specific system functions and, when integrated, provides a centralized platform for managing service transactions, monitoring operational performance, controlling inventory, and generating analytical insights. The modular design of the system ensures that business processes are organized into manageable components, allowing the platform to maintain efficiency, scalability, and maintainability.

The Dashboard Module serves as the central monitoring interface of the system. This module provides users with a summarized view of the business's operational activities through visual indicators, charts, and statistical summaries. The dashboard displays information such as the total number of completed services, pending job orders, scheduled service releases, available inventory levels, and sales performance indicators. By presenting real-time operational data, the dashboard allows the owner and authorized staff to quickly assess the current status of business operations and identify areas requiring immediate attention.

The Job Order Management Module is responsible for managing customer service requests and recording service transactions. Staff members use this

module to create new job orders by entering customer information, selected service type, service cost, item details, and expected release date. The module also allows users to update the status of job orders as they progress from initial intake to completion and release. By centralizing job order records in the database, this module improves organization and enables efficient tracking of service requests.

The Service Management Module allows the owner or administrator to manage the list of services offered by the business. Through this module, authorized users can add new service categories, update service descriptions, modify pricing, and maintain accurate service records. This ensures consistency in service transactions and reduces errors during job order creation.

The Released Calendar Module provides a scheduling interface that allows users to monitor service completion timelines and customer release schedules. This module helps track when services are due for completion or ready for customer pickup. By organizing service schedules in calendar format, staff can monitor deadlines, prioritize workloads, and reduce delays in service delivery.

The Inventory Management Module is responsible for monitoring and controlling the stocks of cleaning materials, restoration supplies, and other consumables used in daily operations. This module allows authorized users to record stock-in and stock-out transactions, update available quantities, and track inventory usage based on completed services. It also provides low-stock alerts whenever supplies reach critical levels. By automating inventory control, the module helps prevent shortages, overstocking, and inaccurate manual stock records.

The Sales Reporting Module generates reports related to service transactions and business performance. This module compiles operational data and produces summaries of daily, weekly, and monthly sales activities. Reports may include revenue totals, service demand trends, top-performing services,

inventory usage summaries, and transaction frequency. These reports help the owner evaluate business performance and support strategic planning.

The User Management Module allows the owner to manage system users and control access privileges. Through this module, user accounts can be created for staff members, login credentials can be managed, and roles can be assigned based on responsibilities. Role-based access control ensures that users can only access modules relevant to their assigned duties. This module enhances system security by preventing unauthorized access to sensitive operational data.

The system also includes a Data Analytics Module that uses historical transaction and service data to generate business insights. Through machine learning algorithms, this module can identify service trends, forecast demand, and support decision-making based on patterns in operational data.

By integrating these modules, the proposed system provides a centralized and efficient platform for service management, inventory control, accurate record keeping, and operational monitoring. The modular architecture enables the business to streamline workflows, reduce manual processes, improve service coordination, and support data-driven decision-making for Shoelotskey.

Process Modeling and System Workflow

Context Diagram

The diagram illustrates the interaction between the system called Shoelotskey, the owner, and the staff, showing how data flows within the system to support business operations.

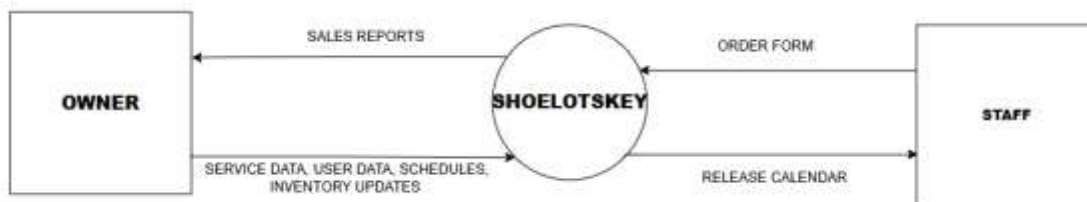


Figure 3: Context Diagram

The owner interacts with the system by providing important operational inputs such as service data, user data, schedules, and inventory updates. These inputs are essential for managing services, maintaining employee records, organizing daily operations, and monitoring stock availability. In return, the system processes this information and generates sales reports for the owner, enabling them to evaluate business performance, monitor revenue, and make strategic decisions for continuous improvement.

On the right side, the staff communicates with the system by submitting order forms, which contain customer job orders, service requests, and transaction details that need to be processed. After receiving these inputs, the system responds by providing a release calendar, which serves as a schedule for completed orders and expected release dates. This helps staff organize workloads, prioritize tasks, and ensure timely service delivery to customers. Overall, the diagram shows that SHOELOTSKEY functions as a centralized information system that streamlines communication and coordination between management and employees. By efficiently receiving inputs and generating relevant outputs, the system supports smoother workflows, better scheduling, improved inventory monitoring, accurate reporting, and more effective management of the shoe service business.

Data Flow Diagram (Level 0)

The diagram illustrates a more detailed data flow of the Shoelotskey system, highlighting how the owner and staff interact with the system to manage operations efficiently.

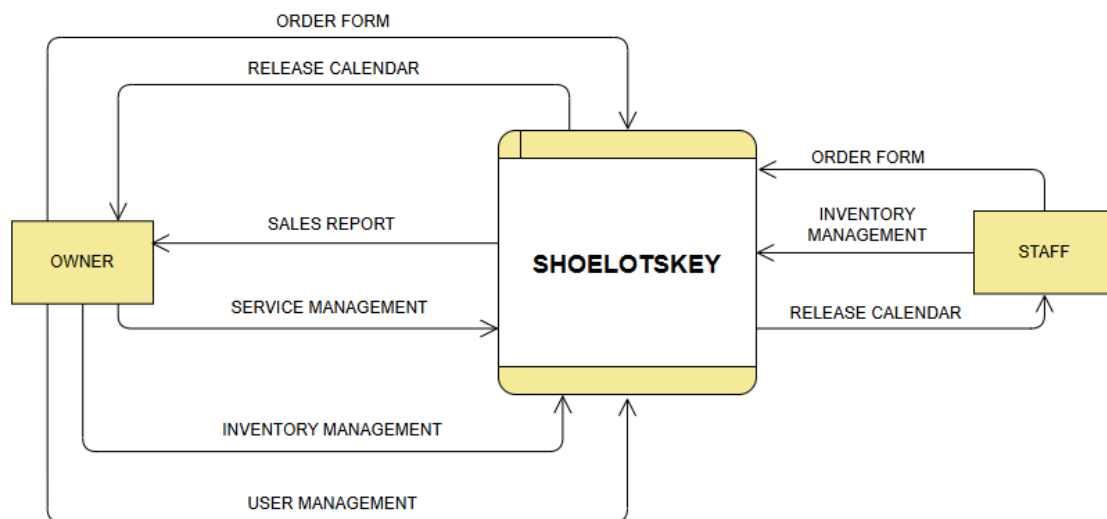


Figure 4: Data Flow Diagram Level 0

The owner performs a major administrative role by interacting with the system through several management functions, including service management, inventory management, and user management. Through service management, the owner can configure available shoe services, pricing, and operational processes. Through inventory management, the owner can monitor stock levels, update materials, and ensure that resources needed for shoe services are always available. Through user management, the owner can maintain employee accounts, assign access privileges, and organize customer or staff records. In addition, the owner receives sales reports generated by the system, which provide valuable insights into revenue, service demand, and business performance for decision-making purposes.

On the other hand, the staff interacts with the system by submitting order forms, which represent customer service requests, job orders, or transaction details that need to be processed. Staff members also provide inventory-related updates to keep stock records accurate and current. After processing these inputs, the system provides a release

Overall, the diagram emphasizes that SHOELOTSKEY serves as a centralized and integrated platform where all operational data from the owner and staff are collected, processed, and transformed into useful outputs such as reports and schedules. By streamlining communication, coordinating tasks, and automating management functions, the system improves workflow efficiency, enhances decision-making, and supports the smooth operation of the shoe service business.

The Level 1 Data Flow Diagram (DFD) illustrates the internal structure and major functional components of the proposed Web-Based Service Management System by breaking down the overall system into detailed processes, data flows, and data stores.



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available to the Owner is job order management, which includes viewing order status summaries, modifying order details, selecting release dates, displaying the release calendar, and reviewing claimed order records. These processes interact with the Orders data store (D1-Orders), where all job order information is stored and updated. By maintaining accurate order records, the system enables the Owner to monitor business transactions, track service progress, and make informed operational decisions.

Another important module shown in the diagram is service management. Through this process, the Owner can modify services, view the list of available services, and display service details. These activities are connected to the Services data store (D2-Services), which contains all information related to shoe services offered by the business, such as cleaning, restoration, repair, and pricing details. Maintaining updated service information ensures consistency in pricing, service availability, and customer expectations, which contributes to better customer satisfaction and smoother operations.

The user management module allows the Owner to control system accounts and personnel records. Through this process, the Owner can display the user list, view user details, and modify user information when necessary. These transactions are connected to the Users data store (D3-Users), where staff account credentials and user records are maintained. This module is essential for controlling access privileges, assigning responsibilities, and ensuring secure use of the system by authorized personnel only.

The diagram also includes expense management, which supports financial tracking of business costs. Through this process, the Owner can display expense records and modify expense details whenever new operational costs are incurred. All expense-related information is stored in the Expenses data store (D4-Expenses). Recording expenses inside the system helps management monitor spending patterns, evaluate profitability, and maintain transparency in financial transactions.

Another critical component is payment management, which handles customer payment transactions for completed services. Payment records such as payment amount, references, and status are stored in the Payments data store (D5-Payments). This ensures that all customer payments are accurately documented and linked to their respective job orders. Proper payment monitoring helps reduce errors, supports financial reporting, and verifies whether orders are fully paid before release or claiming.

The system also emphasizes inventory management, where the Owner can manage inventory records, update item details, and monitor available stocks of materials used for shoe services. These transactions are connected to the Inventory data store (D6-Inventory). This module is highly important because it ensures that cleaning materials, repair supplies, paints, and other resources remain available whenever needed. It also helps prevent shortages and supports better purchasing decisions.

On the operational side, the Staff interacts with the system mainly for service execution and scheduling purposes. Staff members can view order status summaries, modify order details when authorized, select release dates, display the release calendar, and review claim records. They also receive inventory details needed for service completion. These interactions allow staff to stay informed about assigned tasks, monitor deadlines, and organize daily workloads more efficiently. As a result, coordination between management and staff is improved, leading to faster service completion and better customer service.

Finally, the report generation process consolidates data from multiple data stores such as Orders, Expenses, Payments, and Inventory to create financial and business summary reports. These reports provide the Owner with useful insights regarding sales performance, revenue trends, expenses, inventory usage, and overall operational efficiency. By analyzing these reports, management can make strategic decisions that support business growth and continuous improvement.

Overall, the Level 1 DFD demonstrates that SHOELOTSKEY is a fully integrated management information system that combines administrative, operational, financial, and inventory functions into one centralized platform. It ensures smooth data flow, accurate record management, improved staff coordination, and informed decision-making. This detailed process structure highlights how the proposed capstone system can significantly modernize and enhance the management of a shoe service business.

Procedural Workflow

The Procedural Workflow Diagram illustrates the step-by-step workflow of the Shoelotskey system, detailing how a customer service request is processed from start to finish.

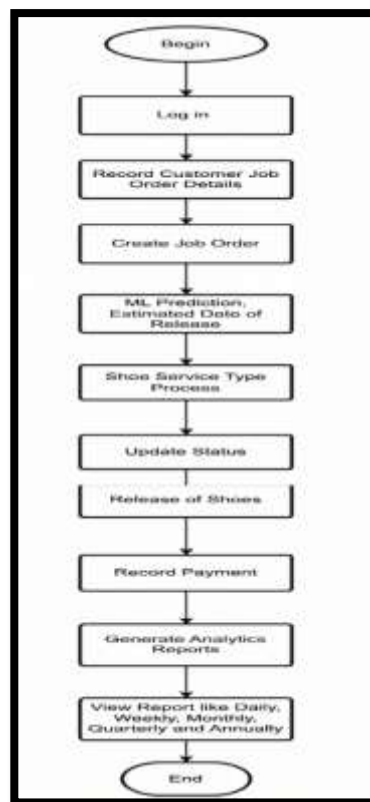


Figure 6: Procedural Workflow

The process begins with the user logging into the system, ensuring secure access before any operation is performed. Once logged in, the staff records the customer's job order details, including the type of service required and relevant

shoe information, which is then used to create a formal job order within the system.

After the job order is created, a machine learning component analyzes historical and current data to predict the estimated date of release, providing both staff and customers with a reliable timeframe. The process then proceeds to the actual shoe service stage, where the requested service such as cleaning or repair is performed. As the work progresses, the system is updated with the current status of the job to maintain accurate tracking. Upon completion, the shoes are released to the customer, followed by the recording of payment to ensure proper financial documentation. The system then generates analytics and reports based on the transaction and operational data, which can be reviewed in various formats such as daily, weekly, quarterly, and annual reports. The workflow concludes with the end of the process, emphasizing a structured, efficient, and data-driven approach to managing shoe service operations.

customers, staff assignments, resources, and predictive outputs. A Customer may submit multiple service requests, establishing a one-to-many relationship that allows the system to store and monitor the transaction history of each client. Customer records include attributes such as customer ID, name, contact information, and customer type, enabling better customer profiling and service management.

The model also includes the Service entity, which stores information about the various shoe services offered by the business, such as cleaning, restoration, repainting, sole repair, and other maintenance services. Each service record contains attributes such as service ID, service name, description, estimated cost, and estimated duration. These services are linked to service requests and are assigned to staff members based on operational needs. The Staff entity contains details such as staff ID, staff name, contact information, role or position, skill set, and availability. Through the Service Assignment relationship, the system can allocate specific service requests to qualified staff members while recording assignment dates, start times, and completion dates. This structure helps improve workforce scheduling, accountability, and monitoring of service progress.

To support operational resources, the model includes the Resource and Resource Allocation entities. The Resource entity stores materials and supplies needed for service operations, with attributes such as resource ID, resource name, description, quantity in stock, and cost. Through the Resource Allocation entity, the system records how many resources are used for each service request, including allocation date and quantity used. This relationship enables accurate inventory tracking, cost monitoring, and resource planning. In addition, the Inventory entity manages stock items such as cleaning materials, paints, adhesives, and repair tools. It includes attributes like item ID, item name, category, stock quantity, unit, and unit price. By linking inventory to services through quantity usage relationships, the system ensures that materials consumed during operations are properly deducted and monitored.

A key innovation of the model is the integration of Machine Learning through the Machine Learning Model and Model Prediction entities. The Machine Learning Model entity stores model-related information such as model ID, model name, description, type, target variable, and input features. These models analyze historical transaction and service data to generate intelligent forecasts. The resulting outputs are stored in the Model Prediction entity, which contains model prediction ID, model ID, service request ID, predicted value or classification, confidence score, and prediction date/time. These predictions can be used to estimate service completion dates, forecast workload demand, or identify service trends. This predictive capability supports faster decision-making, better scheduling, and improved customer satisfaction.

Overall, the conceptual data model ensures that SHOELOTSKEY has a well-structured and organized data foundation capable of supporting both operational and analytical requirements. It integrates customer management, service processing, staff coordination, inventory control, resource monitoring, and predictive analytics into a unified framework. By clearly defining relationships among entities, the model promotes data consistency, operational efficiency, scalability, and intelligent service management, making it highly suitable for the objectives of the proposed capstone project.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) of the Shoelotskey system illustrates the structured relationships among the various entities that support the overall functionality of the shoe service management application.

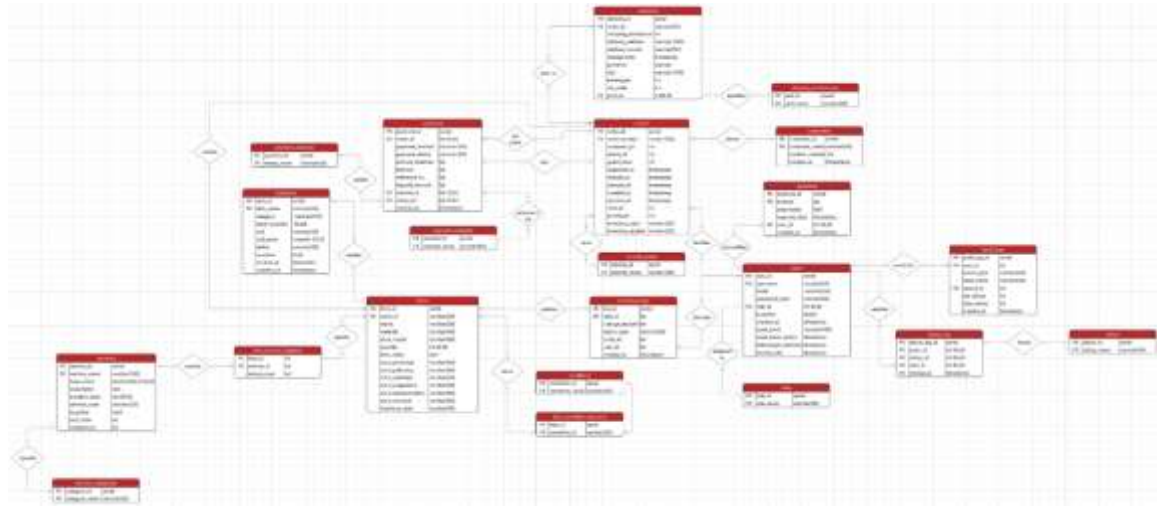


Figure 8: Entity Relationship Diagram

The ERD is centered around the orders table, which serves as the primary transaction entity of the system. This table stores all customer service requests, including order number, customer reference, current status, total amount, expected completion date, release date, assigned user, priority level, and inventory approval. Each service transaction begins when a customer places an order, making this entity the core component that links customer records, items, payments, delivery information, and service status updates.

The customers table stores customer details such as customer name, contact number, and registration date. Each customer can place multiple orders, establishing a one-to-many relationship between customers and orders. This allows the system to maintain customer transaction history and repeat service records.

The items table contains the detailed list of customer footwear items included in an order. Since one order may consist of multiple shoes or service items, the relationship between orders and items is one-to-many. This table

records important item-specific data such as brand, material, shoe model, quantity, notes, scratches, yellowing condition, sole separation, worn-out status, and inventory used. This enables detailed service inspection and customized treatment for each item.

The services table contains the list of available services offered by Shoelotskey, such as basic cleaning, unyellowing, minor retouch, minor restoration, reglue, and color renewal. Each service includes a service name, base price, duration, description, service code, activation status, and category reference. Services are grouped using the service_categories table, allowing better organization of cleaning, repair, restoration, and customization services.

Because one item may require multiple services and one service may apply to multiple items, the ERD uses the item_service_mapping table as a junction table to resolve the many-to-many relationship between items and services. This table stores the selected services applied to each shoe item and may include the actual charged price.

The conditions table stores predefined shoe condition types, such as scratches, yellowing, sole separation, worn-out soles, and discoloration. Since one item may have multiple conditions, the item_condition_mapping table is used as another junction table that links items with their observed conditions during intake assessment.

The payments table records all payment transactions related to customer orders. It stores payment method, payment status, amount received, balance, deposit amount, refund amount, payment method ID, and payment status ID. This table is connected to the payment_methods table, which defines accepted payment options such as cash, GCash, bank transfer, or other methods. It also connects to the payment_statuses table to classify payments as paid, partial, unpaid, refunded, or pending.

The deliveries table stores information for orders requiring delivery or pickup arrangements. It includes shipping preference, delivery address, courier, release time, province, city, barangay, zip code, and shipping fee reference. This entity connects to the shipping_preferences table, which defines available delivery methods such as branch pickup, same-day delivery, courier shipping, or meet-up arrangements.

The users table stores all system user accounts, including username, email, hashed password, assigned role, activation status, reset token, failed login attempts, and lockout settings. This ensures secure system access. The roles table defines user privileges such as owner, administrator, cashier, staff, or technician. This supports Role-Based Access Control (RBAC), ensuring users can only access functions based on their responsibilities.

The status table stores the possible operational states of orders, such as pending, processing, cleaning, restoring, ready for release, released, or cancelled. Order status movements are recorded in the status_log table, which tracks status changes over time, including who updated the status and when the update occurred. This provides accountability and workflow transparency.

The inventory table stores stocks of cleaning materials, paint supplies, adhesives, tools, and consumables used during service operations. It records item name, category, stock quantity, unit, unit price, status, and activation flag. Every inventory movement is recorded in the inventory_logs table, including stock changes, adjustments, order usage, responsible user, and timestamp. This supports accurate stock control and auditability.

The priority_levels table classifies job urgency, such as normal, urgent, express, or premium handling. Orders may be assigned priority levels to help schedule work efficiently and manage customer expectations.

The expenses table records business expenses such as material purchases, utilities, wages, or miscellaneous operational costs. This helps management monitor expenditures and compare them against sales performance.

The audit_logs table provides system-wide activity monitoring by recording user actions such as inserts, updates, deletions, accessed tables, old values, new values, and timestamps. This improves accountability, security monitoring, and incident investigation.

Logical Database Design

The logical database design of the Shoelotskey system presents a detailed and structured representation of how data is organized, related, and managed within the application to support its full range of operations.

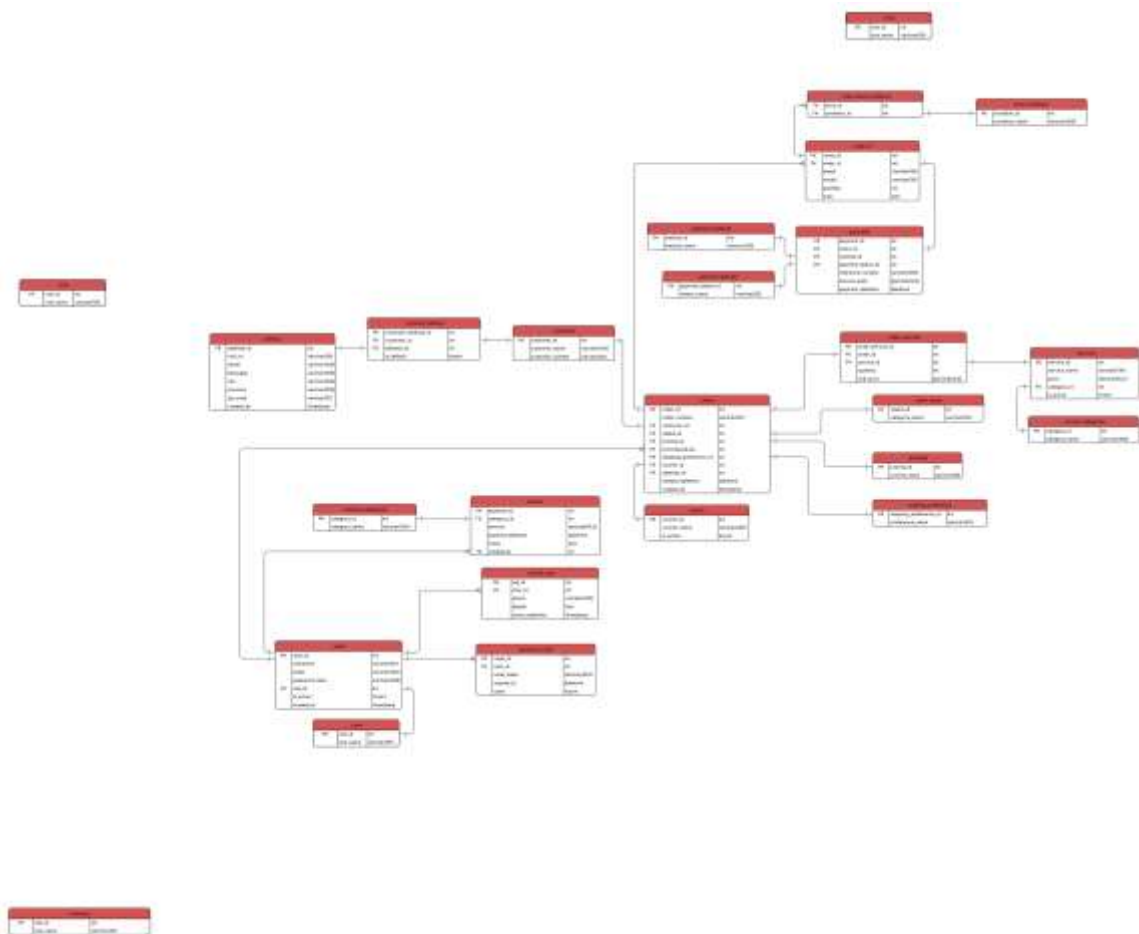


Figure 9: Logical Database Design

At the center of the design is the orders table, which functions as the primary transaction table of the system. It stores important job order details such as order number, customer ID, status ID, priority ID, shipping preference ID, courier ID, address ID, release date and time, and creation timestamp. Each order is linked to a specific customer and can contain one or more shoe items that require service. This structure allows the system to manage multiple customer transactions while maintaining complete records of each job order from creation to completion. The inclusion of order statuses and priorities helps the business monitor progress and organize workloads effectively.

Customer information is managed through the customers table, which stores customer IDs, names, and contact numbers. To support multiple addresses per customer, the design includes an address table and a customer_address linking table. The address table contains detailed location fields such as unit number, street, barangay, city, province, and zip code. Through this relational setup, customers may have several saved addresses while one can be marked as the default address. This improves convenience for repeat customers and supports efficient delivery or pickup coordination.

The database also handles shoe item details through the order_id table, which records each shoe associated with an order. Attributes include brand, model, quantity, and notes. To document shoe condition before service, the system includes a shoe_conditions table and an order_shoe_conditions linking table. These tables allow staff to record visible conditions such as scratches, sole damage, stains, or other issues before processing. This feature is important for service transparency, proper assessment, and customer trust.

Payment transactions are managed through the payments table, which stores payment ID, order ID, payment method ID, payment status ID, reference number, amount paid, and payment date/time. Supporting tables such as payment_methods and payment_statuses define accepted payment channels and payment states such as paid, partial, or pending. This structure ensures accurate

recording of customer payments and enables the business to monitor outstanding balances and completed transactions.

Service management is handled through the services table, which stores the list of shoe services offered by the business. Fields include service name, price, category ID, and active status. Examples of services may include cleaning, repainting, restoration, sole repair, and deodorizing. Related to this is the service_categories table, which organizes services into categories for easier management and filtering. Since one order may require multiple services, the system uses the order_services table as a junction table that links orders to selected services while recording quantity and unit price. This many-to-many relationship provides flexibility in processing customized customer requests.

The database design also supports order monitoring through the order_status and priorities tables. The order_status table stores possible statuses such as pending, in progress, completed, released, or claimed. Meanwhile, the priorities table defines urgency levels that help staff identify rush orders or important tasks. In addition, the shipping_preference and courier tables allow the system to record whether the customer prefers pickup, delivery, or a specific courier service. These tables improve order tracking and release coordination.

Administrative and security functions are managed through the users table, which stores user ID, username, email, password hash, role ID, active status, and account creation date. User roles are defined in the roles table, allowing the system to distinguish between owner, administrator, and staff access levels. To enhance security, the design also includes a password_reset table for password recovery requests and an activity_logs table for recording user actions within the system. These tables help ensure accountability, traceability, and secure access management.

Financial monitoring is further supported through the amount table, which records operational expenses such as category ID, amount, expense date/time, notes, and creator ID. Expense categories are maintained in the

expense_categories table, allowing classification of costs such as supplies, utilities, transportation, or maintenance. This module enables the owner to track business expenditures and compare them against revenues for profitability analysis.

Overall, the Logical Database Design demonstrates that SHOELOTSKEY uses a comprehensive and well-normalized relational database capable of handling both front-end operations and back-end administration. By integrating customer management, order processing, shoe condition tracking, service selection, payments, logistics, user security, and expense monitoring into one database structure, the system ensures reliable data storage, faster transaction processing, and accurate reporting. This design supports scalability, maintainability, and long-term efficiency, making it highly suitable for the successful implementation of the proposed capstone project.

System Security and Data Privacy Design

The System Security and Data Privacy Design of the proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithms for Shoelotskey in Barangay 183 Villamor–Pasay aims to ensure the protection, confidentiality, and integrity of operational data stored and processed within the system. Since the system manages sensitive business information, including job orders, service transactions, customer information, and user credentials, appropriate security measures are implemented to prevent unauthorized access, data breaches, and misuse of the system.

The system's security framework integrates multiple protection mechanisms, including authentication controls, role-based access management, encryption protocols, and system activity monitoring. These mechanisms collectively ensure that only authorized users can access system functionality while protecting operational data from unauthorized modification or disclosure.

In addition to technical security measures, the system also incorporates responsible data privacy practices that regulate how information is collected, stored, and processed within the platform. Proper data handling procedures are implemented to ensure that operational information is used only for legitimate service management purposes. These security and privacy mechanisms enhance system reliability while ensuring that Shoelotskey maintains responsible management of business and customer data.

Access Control and Authentication Mechanisms

The system implements secure authentication and role-based access control mechanisms to regulate user access to system functionalities. Authentication ensures that users must verify their identity before accessing the platform, thereby preventing unauthorized access to system resources.

Each user is required to log in to the system using a unique username and password assigned during account creation. The authentication process verifies the provided credentials against the stored user records within the system database. Only users with valid credentials are granted access to the system interface.

The system supports two primary user roles:

Owner– The owner functions as the primary system administrator and decision-maker. This role has full access to system modules, including dashboard monitoring, service management, job order review, report generation, and user account administration. The owner can also analyze business performance using data analytics generated by the system.

Staff– Staff users are responsible for managing day-to-day service operations. Their responsibilities include recording customer service requests, creating job orders, updating service status, and monitoring service completion schedules.

Role-based access control ensures that users can only access functions that correspond to their assigned responsibilities. This approach enhances system security while maintaining operational efficiency by preventing unauthorized users from modifying sensitive system information.

Data Encryption and Protection Measures

The proposed system incorporates comprehensive data encryption and protection measures to safeguard sensitive operational information stored in the database and transmitted across network connections. These security mechanisms are designed to maintain the confidentiality, integrity, and availability of business data throughout the operation of the system. To secure communication between users and the web server, encrypted transmission protocols are implemented to prevent unauthorized interception of sensitive information such as login credentials, customer details, and transaction records while data is being transferred over the network.

Sensitive operational data, including job orders, service records, inventory information, transaction history, and user credentials, are stored in a centralized relational database with controlled access permissions. Only authorized users and validated system processes are granted access to view, modify, or manage records. This helps prevent unauthorized access, accidental modification, and misuse of critical business information. In addition, the system implements regular backup and recovery procedures to protect data from accidental deletion, hardware failure, or security incidents. Periodic database backups ensure that records can be restored quickly in case of unexpected system disruptions.

To further strengthen security, the system follows OWASP Top 10 hardening practices that address common web application vulnerabilities. Broken access control is mitigated through JWT token validation and Role-Based Access Control (RBAC), ensuring that only authorized users can access specific system functions based on their roles. Cryptographic failures are prevented by securing passwords using Bcrypt hashing with a 12-round salt configuration. Injection

attacks, such as SQL injection, are mitigated through the use of parameterized queries implemented via SQLAlchemy ORM.

The system also addresses insecure design by applying Third Normal Form (3NF) relational database constraints to preserve data consistency and integrity. Security misconfiguration risks are reduced through the use of hardened HTTP security headers, restricted CORS policies, and environment-based credential management. To minimize risks from vulnerable or outdated components, continuous dependency auditing is performed on major frameworks and libraries such as FastAPI, Pydantic, and React. Identification and authentication failures are mitigated through multi-layer token validation and automated session expiration controls.

Moreover, the system protects software and data integrity through a real-time monitoring component developed in Node.js that detects unauthorized file modifications or suspicious changes within the environment. Logging and monitoring failures are addressed through centralized real-time logs of API requests, user actions, and data modifications for auditing and incident response purposes. Lastly, server-side request forgery (SSRF) attacks are mitigated by restricting and validating outgoing server-side requests to prevent unauthorized communication with external resources.

Through these integrated security measures, the proposed system ensures reliable protection of operational data while providing a secure and stable environment for managing the daily business processes of Shoelotskey.

Data Privacy and Ethical Considerations

The proposed system is designed to follow responsible data privacy and ethical data handling practices to protect the privacy of individuals whose information may be recorded during service transactions. Customer information and service records are treated as confidential data and are stored securely within the system database.

Access to operational data is restricted only to authorized users based on their assigned roles within the system. This restriction ensures that sensitive information cannot be accessed, shared, or modified by unauthorized individuals.

All data collected by the system is used exclusively for service management, operational monitoring, and business reporting purposes. The system does not allow unauthorized disclosure or misuse of stored information. Ethical considerations in data management ensure that information is handled responsibly while supporting transparency and accountability in business operations.

These privacy and ethical safeguards strengthen user trust in the system and ensure responsible management of business data within the Shoelotskey service environment.

Audit Logging and Activity Monitoring

To maintain transparency and accountability within system operations, the proposed platform incorporates audit logging and activity monitoring mechanisms. Audit logs record significant system activities performed by users and system processes.

The system logs events such as user authentication attempts, job order creation, service status updates, report generation, and user account modifications. Each log entry includes relevant details, such as the user responsible for the action, the activity performed, and the event timestamp.

Audit logs provide a reliable record of system usage and allow administrators to track operational activities within the platform. These logs can also be used to investigate potential security incidents or unusual system behavior.

Activity monitoring tools enable administrators to detect unauthorized access attempts, abnormal system activity, or potential misuse of system

resources. By maintaining detailed activity records, the system improves operational transparency while strengthening system security.

Implementation Environment and Technology Stack

The implementation environment and technology stack describe the technological infrastructure used to develop, deploy, and operate the proposed web-based system. The system is built using modern web technologies that support efficient application development, secure data management, and reliable system performance.

The technology stack integrates server-side programming frameworks, relational database systems, and client-side interface technologies to enable the system to function as a fully web-based platform accessible through internet browsers. This architecture allows users to interact with the system through a graphical interface while the application server processes system logic and communicates with the database server to retrieve or store operational data.

Software Requirements

The software requirements specify the system components needed to develop and operate the web-based service management platform.

The system requires a web server platform capable of hosting the application and processing user requests. Server-side software handles system logic, data processing, and machine-learning analytics to analyze historical service records.

A relational database management system is required to store and manage system data, including job orders, service records, transaction details, and user account information. Users access the system through modern web browsers, which provide an interface for interacting with system modules such as dashboards, job order forms, service management pages, and sales reports.

These software components collectively provide the necessary environment for operating the web-based service management system efficiently and securely.

Hardware Requirements

The hardware requirements for the proposed system include computing devices capable of accessing web-based applications and supporting internet connectivity.

The owner and staff members will use desktop or laptop computers equipped with modern web browsers to interact with the system interface. These devices allow users to record service transactions, update job orders, and monitor business operations.

The system will be hosted on a server environment capable of handling application processing, database management, and network communication. The server infrastructure does not require specialized hardware, making the system suitable for small- and medium-scale business environments.

Stable internet connectivity is required to enable communication between users and the web server, ensuring smooth system access and data synchronization.

Development Tools and Programming Technologies

The Web-Based Service Management System for Shoelotskey is developed using a modern and reliable technology stack suited for high-performance web applications.

The frontend of the system is built using React with TypeScript to ensure structured and maintainable code. Vite is used to support fast development and efficient system performance, while Tailwind CSS provides a clean, responsive,

and user-friendly interface design. Lucide React is integrated to maintain a consistent and professional set of icons across the system.

The backend is developed using Python with the FastAPI framework. This setup allows the system to handle multiple user requests efficiently while enforcing business rules and processing system logic. FastAPI also supports smooth integration with mathematical and analytical components.

For data storage, the system uses a PostgreSQL relational database. SQLAlchemy ORM is applied to manage database operations, ensuring data accuracy, consistency, and integrity throughout the system.

The system's predictive analytics feature is implemented using the Scikit-Learn library. A Random Forest regression model is used to estimate service completion timelines based on historical service data, helping improve operational planning and decision-making.

To ensure security, user passwords are protected using the Bcrypt hashing algorithm, preventing unauthorized access and enhancing overall system protection.

Overall, this technology stack ensures the system is secure, scalable, efficient, and suitable for real-world business operations, making it appropriate for a capstone-level implementation.

Deployment Environment

The proposed system is deployed within a secure, cloud-based web environment to ensure wide accessibility, reliability, and efficient system performance. As a web-based application, it enables authorized users such as the business owner and staff to access the platform through standard internet browsers using either the organization's local network or an internet connection. The deployment process involves installing the application on a secure web server and configuring a centralized database server to manage operational data

effectively. This architecture supports seamless communication between system components while maintaining data integrity and accessibility.

For production, the system is hosted on the Heroku cloud platform, where both frontend and backend components are deployed using dedicated dynos to ensure stable and efficient execution. The application's database is managed Heroku Postgres, which provides high availability, automated backups, and secure database connections. Additionally, the system integrates the Mailgun API to handle notifications and administrative communications, including transactional emails such as password reset requests and system alerts.

To ensure data security and confidentiality, the entire system is protected by SSL/TLS encryption, safeguarding all data transmitted between users and the server from unauthorized access and potential tampering. The deployment environment also incorporates regular system monitoring, maintenance procedures, and software updates to maintain optimal performance and address potential vulnerabilities. Overall, this deployment strategy delivers a secure, reliable, and scalable platform that effectively supports service operations and informed decision-making for Shoelotskey, making it suitable for real-world implementation in a capstone-level project.

CHAPTER 4

Research Methodology, Results, and Discussions

Research Design

This study used a developmental-descriptive research design. It focused on the design, development, and evaluation of a Web-Based Service Management System with Data Analytics using Machine Learning Algorithms. The developmental aspect was used to create the proposed system, while the descriptive aspect was applied to assess its functionality, usability, and relevance to the business operations. The study followed the Waterfall Software Development Life Cycle (SDLC), which includes requirements gathering, system design, development, testing, deployment, and maintenance.

Data Gathering Technique

The researchers gathered data through interviews, focus group discussions, direct observation, document review, system testing, and evaluation survey questionnaires. All data gathering activities were conducted with the permission of the business owner and with the voluntary participation of respondents. Participants were informed of the purpose of the study, and all collected information was used strictly for academic and system development purposes only. Confidential business records and personal information were treated with privacy and were not disclosed to unauthorized individuals.

Interviews were conducted with one (1) business owner and five (4) staff members of Shoelotskey from March 5 to March 10, 2025 to identify current problems, operational challenges, and system requirements. Each interview lasted approximately 20 to 30 minutes using a semi-structured interview guide. The responses were documented and analyzed to determine the features needed in the proposed system.

A focus group discussion involving the same six (5) participants was conducted on March 12, 2025 to allow respondents to collectively share their experiences, concerns, and suggestions regarding the existing manual process and the features needed in the proposed system. The discussion was facilitated by the researchers and recorded through written notes for analysis.

Direct observation was conducted from March 13 to March 15, 2025 during regular business hours to examine the actual workflow, service transactions, customer handling procedures, and record-keeping practices of the business. Observation helped validate the information gathered during the interviews and discussions.

Document review was also applied from March 16 to March 20, 2025 to analyze existing job orders, service records, inventory logs, and sales reports used as references in designing the system. Approximately 500 historical completed transaction records covering the period of January 2024 to April 2025 were reviewed and used as the original dataset for the machine learning component. Incomplete, duplicate, and inconsistent records were excluded during data cleaning to maintain data integrity and reliability.

After the proposed system was developed, system testing was conducted from June 1 to June 10, 2025 with the participation of the owner and staff to verify functionality, usability, and accuracy of the developed modules. This included testing of login functions, transaction encoding, inventory updates, report generation, and prediction outputs.

Finally, evaluation questionnaires were distributed to the same six (5) respondents from June 15 to June 18, 2025 to assess the acceptability, usefulness, and ease of use of the developed system using a 5-point Likert scale questionnaire based on ISO/IEC 25010 and Technology Acceptance Model (TAM) criteria.

To address ethical considerations, the identities of respondents and confidential business information were kept private and were not disclosed in any

part of the study. Participation was voluntary, and respondents were free to decline answering any question not related to the study. No fabricated, manipulated, or externally sourced data were used. All gathered data were obtained directly from Shoelotskey's authorized records and actual respondents, and all information was processed honestly, objectively, and responsibly to preserve the credibility and ethical integrity of the research findings.

Statistical Treatment of Data

The data gathered from the evaluation of the proposed system were analyzed using descriptive statistics, particularly the weighted mean, to determine the level of acceptability of the system in terms of functionality, usability, reliability, efficiency, and security. The weighted mean was used to summarize and interpret the responses from the respondents based on the evaluation questionnaire. In addition, the study applied regression analysis as part of the machine learning component to develop a service completion date prediction model. Historical service data were processed using Python with scikit-learn to identify patterns and generate predictions regarding expected service completion dates. The predictive performance of the model was evaluated using Mean Absolute Error (MAE) to measure the average difference between the actual and predicted completion dates. A lower MAE value indicated better predictive accuracy and model reliability.

Formula:

$$WM = \frac{\sum fx}{N}$$

Where:

WM = Weighted Mean

f = frequency of responses

x = weight of each response

N = total number of respondents

Population and Sampling Technique

The respondents of this study will consist of individuals directly involved in the operation and evaluation of the proposed system for Shoelotskey. The target population includes the business owner and five employees who are responsible for customer transactions, service processing, shoe cleaning and restoration services, and inventory-related tasks. These respondents are considered essential because they are the intended end-users of the developed web-based service management system and can provide firsthand feedback regarding its usability, functionality, and overall acceptance.

In addition to the internal respondents, the study will include selected Information Technology (IT) experts to evaluate the technical quality of the system based on ISO/IEC 25010 standards. These experts may include IT instructors, software developers, system analysts, programmers, or professionals with experience in software evaluation and system development. Their participation is necessary to assess the system in terms of functional suitability, performance efficiency, reliability, security, maintainability, usability, and portability.

The study will use total population sampling for the Shoelotskey staff and management since the number of internal respondents is small and manageable. This allows all employees and management personnel to participate in evaluating the system. For the IT experts, purposive sampling will be used to select qualified evaluators with relevant technical knowledge and experience in software development and quality assessment.

By combining operational users and technical evaluators, the study ensures a comprehensive assessment of the proposed system from both user and expert perspectives. The internal respondents will assess user acceptance using the Technology Acceptance Model (TAM), while the IT experts will evaluate the software quality using ISO/IEC 25010 criteria.